



DEPARTMENT OF REGULATORY AND ECONOMIC RESOURCES (RER)
BOARD AND CODE ADMINISTRATION DIVISION

MIAMI-DADE COUNTY
PRODUCT CONTROL SECTION
11805 SW 26 Street, Room 208
Miami, Florida 33175-2474
T (786) 315-2590 F (786) 315-2599

www.miamidade.gov/economy

NOTICE OF ACCEPTANCE (NOA)

American Windows & Shutters, Inc.
11600 Adelmo Lane, Suite 1
Fort Myers, Fl. 33966

SCOPE:

This NOA is being issued under the applicable rules and regulations governing the use of construction materials. The documentation submitted has been reviewed and accepted by Miami-Dade County RER- Product Control Section to be used in Miami Dade County and other areas where allowed by the Authority Having Jurisdiction (AHJ).

This NOA shall not be valid after the expiration date stated below. The Miami-Dade County Product Control Section (In Miami Dade County) and/or the AHJ (in areas other than Miami Dade County) reserve the right to have this product or material tested for quality assurance purposes. If this product or material fails to perform in the accepted manner, the manufacturer will incur the expense of such testing and the AHJ may immediately revoke, modify, or suspend the use of such product or material within their jurisdiction. RER reserves the right to revoke this acceptance if it is determined by Miami-Dade County Product Control Section that this product or material fails to meet the requirements of the applicable building code.

This product is approved as described herein and has been designed to comply with the High Velocity Hurricane Zone of the Florida Building Code.

DESCRIPTION: ES 55 Extruded Aluminum Roll-Up Shutter

APPROVAL DOCUMENT: Drawing No. 20-060, titled "ES 55 Extruded Aluminum Roll-up Shutter", prepared by PPMF Inc., dated June 21, 2020, sheets 1 through 14 of 14, signed and sealed by Pedro De Figueiredo P.E., on June 21, 2020, bearing the Miami-Dade County Product Control Revision stamp with the Notice of Acceptance number and the expiration date by the Miami-Dade County Product Control Section.

MISSILE IMPACT RATING: Large and Small Missile Impact Resistant

LABELING: Each unit shall bear a permanent label with the manufacturer's name or logo, city, state, the following statement: "Miami-Dade County Product Control Approved", and NOA number, per TAS-201, TAS-202, and TAS-203, unless otherwise noted herein.

RENEWAL of this NOA shall be considered after a renewal application has been filed and there has been no change in the applicable building code negatively affecting the performance of this product.

TERMINATION of this NOA will occur after the expiration date or if there has been a revision or change in the materials, use, and/or manufacture of the product or process. Misuse of this NOA as an endorsement of any product, for sales, advertising or any other purposes shall automatically terminate this NOA. Failure to comply with any section of this NOA shall be cause for termination and removal of NOA.

ADVERTISEMENT: The NOA number preceded by the words Miami-Dade County, Florida, and followed by the expiration date may be displayed in advertising literature. If any portion of the NOA is displayed, then it shall be done in its entirety.

INSPECTION: A copy of this entire NOA shall be provided to the user by the manufacturer or its distributors and shall be available for inspection at the job site at the request of the Building Official.

This NOA revises & renews NOA #20-0630.17 and consists of this page 1, evidence submitted pages E-1, E-2, E-3 & E-4 as well as approval document mentioned above.

The submitted documentation was reviewed by **Helmy A. Makar, P.E., M.S.**



Helmy A. Makar
07/20/2023

NOA No. 23-0525.01
Expiration Date: 07/03/2024
Approval Date: 07/20/2023

American Windows & Shutters, Inc.

NOTICE OF ACCEPTANCE: EVIDENCE SUBMITTED

1. EVIDENCE SUBMITTED UNDER PREVIOUS APPROVAL # 08-0324.13

A. DRAWINGS

1. *Drawing No. 08-049, titled "ES 55 Extruded Aluminum Roll-up Shutter", prepared by EngCo, Inc. dated 02/26/08 sheets 1 through 14 of 14 with no revisions, signed and sealed on 03/14/08 by Pedro De Figueiredo P.E.*

B. TESTS

1. *Fenestration Testing Laboratory, Inc. Test Report # 5448 dated 12/17/07 for testing standard TAS 201, TAS 202 & TAS 203 signed and sealed by Carlos S. Rionda, P.E.*
2. *National Certified Testing Laboratories Test Report # 210-2963-1 dated 12/21/03 for Testing standard TAS 201, TAS 202 and TAS 203, signed and sealed by Gerald J. Ferrara, P.E.*
3. *Construction Testing Corporation Test Report #05-011 dated 05/31/05 For testing Standard TAS 202, signed and sealed by Yamil G. Kuri, P.E.*
4. *Fenestration Testing Laboratory, Inc. Test Report # 5611 dated 05/03/08 for testing standard TAS 201, and TAS 203 signed and sealed by Carlos S. Rionda, P. E.*

C. CALCULATIONS

1. *Structural Calculations and design for Slat, Track Anchorage, Header, Mullion, Storm Bars and analysis of test results prepared by EngCo, Inc. dated 03/07/08 signed and sealed by Pedro De Figueiredo P.E.*

D. MATERIAL CERTIFICATIONS

1. *None.*

E. STATEMENTS

1. *Code Compliance and no financial interest letter by EngCo Inc .signed and sealed on March 07, 2008 by Pedro De Figueiredo, P.E.*
2. *Code Compliance letter for the tests performed by National Certified Laboratories, signed and sealed by Gerald J. Ferrara, P.E.*

F. QUALITY ASSURANCE

1. *Miami-Dade County Building Code Compliance Office.*

2. EVIDENCE SUBMITTED UNDER PREVIOUS APPROVAL # 13-0327.03

A. DRAWINGS

1. *Drawing No. 13-017, titled "ES 55 Extruded Aluminum Roll-up Shutter", prepared by EngCo., Inc., dated January 16, 2013, sheets 1 through 14 of 14 with no revisions, signed and sealed by Pedro De Figueiredo P.E., on March 11, 2013.*



Helmy A. Makar, P.E., M.S.
Product Control Section Supervisor
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B. TESTS

1. *None.*

C. CALCULATIONS

1. *None.*

D. QUALITY ASSURANCE

1. *By Miami-Dade County Department of Regulatory and Economic Resources.*

E. MATERIAL CERTIFICATIONS

1. *None.*

F. STATEMENTS

1. *Code Compliance and no financial interest letter by EngCo., Inc., signed and sealed by Pedro De Figueiredo, P.E., on March 11, 2013.*

3. EVIDENCE SUBMITTED UNDER PREVIOUS APPROVAL # 16-0531.06

A. DRAWINGS

1. *Drawing No. 16-097, titled "ES 55 Extruded Aluminum Roll-up Shutter", prepared by EngCo., Inc., dated April 25, 2016, sheets 1 through 14 of 14, signed and sealed by Pedro De Figueiredo P.E., on May 16, 2016.*

B. TESTS

1. *None.*

C. CALCULATIONS

1. *Structural Calculations, Track Anchorage, prepared by EngCo, Inc. dated 05/16/16 signed and sealed by Pedro De Figueiredo P.E.*

D. QUALITY ASSURANCE

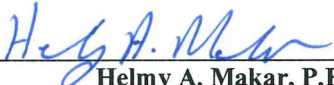
1. *By Miami-Dade County Department of Regulatory and Economic Resources.*

E. MATERIAL CERTIFICATIONS

1. *None.*

F. STATEMENTS

1. *Florida Building Code, 2014 Edition Compliance letter by EngCo., Inc., signed and sealed by Pedro De Figueiredo, P.E., on May 16, 2016.*



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NOTICE OF ACCEPTANCE: EVIDENCE SUBMITTED

4. EVIDENCE SUBMITTED UNDER PREVIOUS APPROVAL # 18-0425.01

A. DRAWINGS

1. *Drawing No. 18-228, titled "ES 55 Extruded Aluminum Roll-up Shutter", prepared by PPMF Inc., dated April 03, 2018, sheets 1 through 14 of 14, signed and sealed by Pedro De Figueiredo P.E., on April 03, 2018.*

B. TESTS

1. *None.*

C. CALCULATIONS

1. *None.*

D. QUALITY ASSURANCE

1. *By Miami-Dade County Department of Regulatory and Economic Resources.*

E. MATERIAL CERTIFICATIONS

1. *None.*

F. STATEMENTS

1. *Florida Building Code, 2017 Edition Compliance letter by PPMF Inc., signed and sealed by Pedro De Figueiredo, P.E., on April 03, 2018.*

5. EVIDENCE SUBMITTED UNDER PREVIOUS APPROVAL # 20-0630.17

A. DRAWINGS

1. *Drawing No. 20-060, titled "ES 55 Extruded Aluminum Roll-up Shutter", prepared by PPMF Inc., dated June 21, 2020, sheets 1 through 14 of 14, signed and sealed by Pedro De Figueiredo P.E., on June 21, 2020.*

B. TESTS

1. *None.*

C. CALCULATIONS

1. *None.*

D. QUALITY ASSURANCE

1. *By Miami-Dade County Department of Regulatory and Economic Resources.*

E. MATERIAL CERTIFICATIONS

1. *None.*

F. STATEMENTS

1. *Florida Building Code, 2017 Edition and 2020 Edition Compliance letter by PPMF Inc., signed and sealed by Pedro De Figueiredo, P.E., on June 21, 2020.*



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NOTICE OF ACCEPTANCE: EVIDENCE SUBMITTED

6. NEW EVIDENCE SUBMITTED

A. DRAWINGS

1. *None.*

B. TESTS

1. *None.*

C. CALCULATIONS

1. *None.*

D. QUALITY ASSURANCE

1. *By Miami-Dade County Department of Regulatory and Economic Resources.*

E. MATERIAL CERTIFICATIONS

1. *None.*

F. STATEMENTS

1. *Florida Building Code, 2020 Edition Compliance letter by PPMF Inc., signed and sealed by Pedro De Figueiredo, P.E., on May 01, 2023.*
2. *Amended and Restated Articles of Incorporation of American Shutter Products, Inc.*



Helmy A. Makar, P.E, M.S.
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GENERAL NOTES:

1- DEFINITION: THIS PRODUCT IS A ROLL-UP TYPE SHUTTER; DESIGNED, CONSTRUCTED AND ERECTED TO EASILY ENCLOSE AN AREA, PROVIDING PROTECTION FROM HIGH VELOCITY HURRICANE FORCE WINDS WITHIN THE ALLOWABLE DESIGNED PRESSURES AND LIMITATIONS STATED IN THIS APPROVAL.

2- CODE: THIS PRODUCT HAS BEEN TESTED UNDER THE TAS 201, 202 AND 203 (LARGE & SMALL MISSILE) STANDARDS AND DESIGNED IN ACCORDANCE WITH THE FLORIDA BUILDING CODE, 2020 - 7th EDITION, HVHZ (HIGH VELOCITY HURRICANE ZONES).

3- POSTING: A LEGIBLE DECAL SHALL BE PLACED AT A READILY VISIBLE LOCATION STATING THE FOLLOWING:

"ES55 ROLL-UP SHUTTER
CROCI NORTH AMERICA - FORT MYERS, FLORIDA
MIAMI-DADE COUNTY PRODUCT APPROVED"

4- LOADS: DESIGNED LOAD CALCULATED BASED ON THE ASCE 7-16 AND PROVIDED BY A PROFESSIONAL ARCHITECT OR ENGINEER FOR EACH SPECIFIC PROJECT. THE CALCULATED DESIGNED PRESSURE MUST NOT EXCEED THE ALLOWABLE PRESSURES HERE IN SPECIFIED. THE DESIGN PRESSURES, AS DETERMINED FROM ASCE 7-16, ARE TO BE MULTIPLE BY 0.6

5- MATERIAL: ALL EXTRUDED ALUMINUM SHAPES SHALL BE MADE OF 6063-T6 OR AS NOTED.

6- FASTENERS: ASSEMBLY SCREWS AND ANCHORS SHALL BE AS SPECIFIED IN THE CURRENT SET OF DRAWINGS. INSTALLATION AND LOADS AS PER THIS APPROVAL. NO 33 1/3% INCREASE IN ALLOWABLE STRESS HAS BEEN USED IN THE DESIGN OF THIS PRODUCT. ANCHORS INTO WOOD WERE DESIGNED WITH 1.6 LOAD DURATION. (Kd=.85)

7- MULTIPLE SPAN SHUTTERS: MULTIPLE SPAN COMPONENTS SUCH AS STORM BARS, MULLIONS AND HEADERS, TO BE DESIGNED AS PER THIS SET OF DRAWINGS.

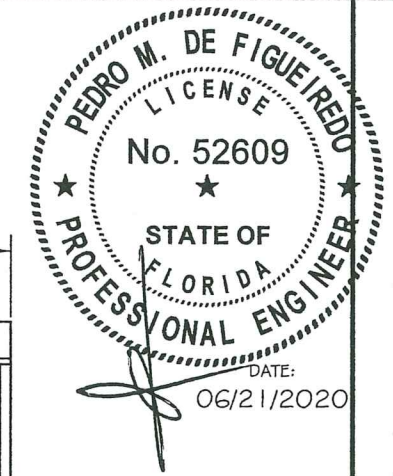
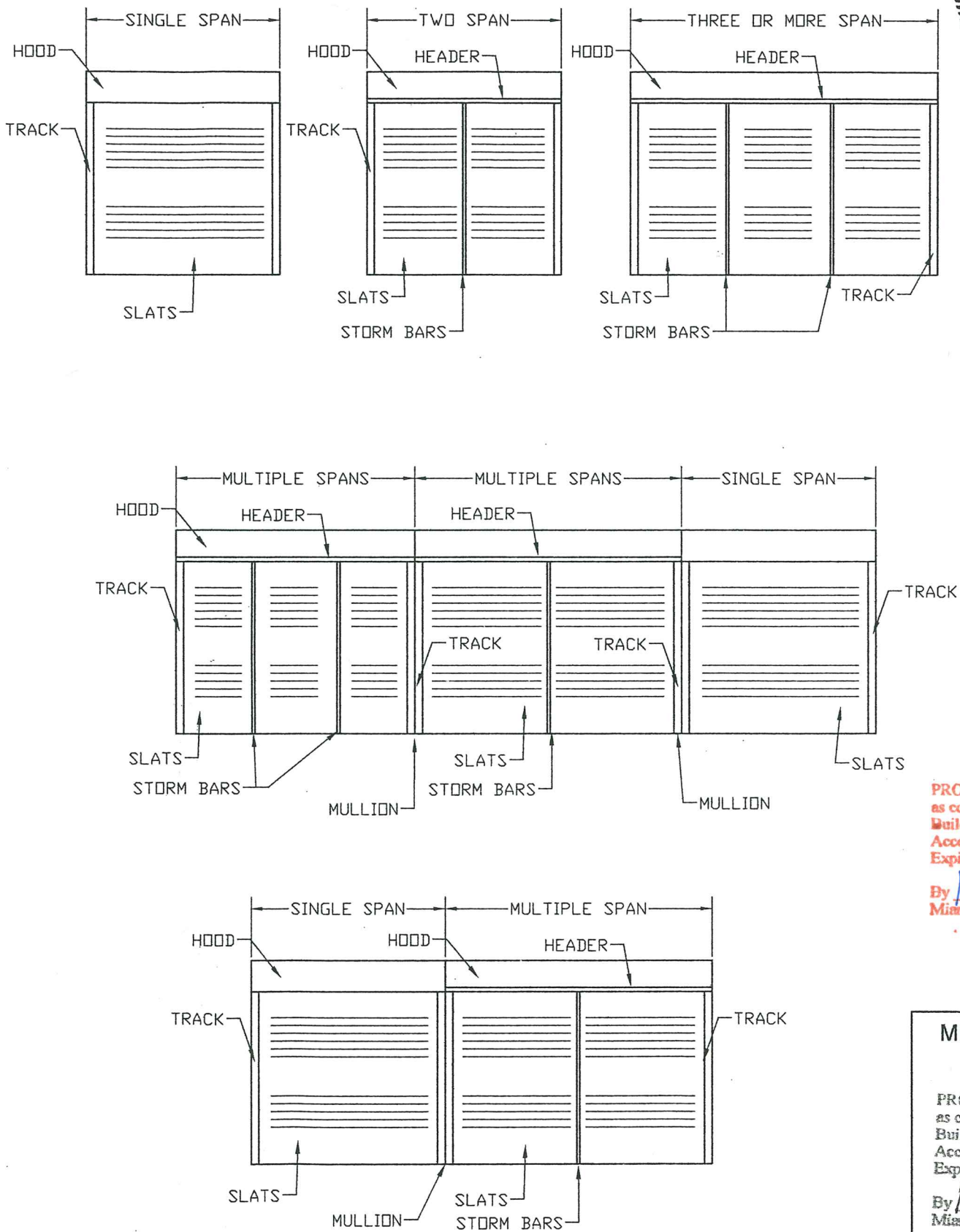
8- USE: IT SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR, ARCHITECT OR ENGINEER OF RECORD TO VERIFY THE FOLLOWING:

- 8.1- THE STABILITY OF THE STRUCTURE WHERE THE SHUTTER IS TO BE ATTACHED INSURING PROPER ANCHORAGE.
- 8.2- THE SITE SPECIFIC PROJECT CRITERIA, SUCH AS BUT NOT LIMITED TO, WIND LOADS, LOCAL CODE REQUIREMENTS, DESIGNED PRESSURES ETC.
- 8.3- THAT THIS APPROVAL IS ADEQUATE TO THE SPECIFIC PROJECT.

9- DISSIMILAR MATERIALS: WHERE ALUMINUM IS IN CONTACT OR FASTENED TO DISSIMILAR MATERIALS, INCLUDING BUT NOT LIMITED TO STEEL, INSTALLATION SHALL MEET THE REQUIREMENTS OF CHAPTER M.7 OF THE ALUMINUM DESIGN MANUAL 2015.

DESIGNED PRESSURE	
COMPONENT	REF. SHEET
SLAT SPAN	3
TRACK ANCHORAGE	4
STORM BARS	5-8
HEADERS	9
MULLIONS	10-14
NOTES: 1- ALLOWABLE DESIGNED PRESSURE FOR MULTIPLE SPANS SHUTTERS ARE FUNCTION OF THE LOWEST ALLOWABLE DESIGNED LOAD FOR THE SLATS, STORM BARS, HEADERS, MULLIONS.	

TYPICAL SHUTTER CONFIGURATIONS



Past & Present Moving the Future
ENGINEERING SERVICES - CA 32459
9715 W. BROWARD BLVD.
PLANTATION, FL 33324
PMF@AOL.COM
ENGINEER SEAL
PEDRO DE FIGUEIREDO
PE 52609

PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

PRODUCT REVISED
as complying with the Florida
Building Code
Acceptance No 23-0525.01
Expiration Date 07/03/2024
By H. G. A. M. M.
Miami Dade Product Control

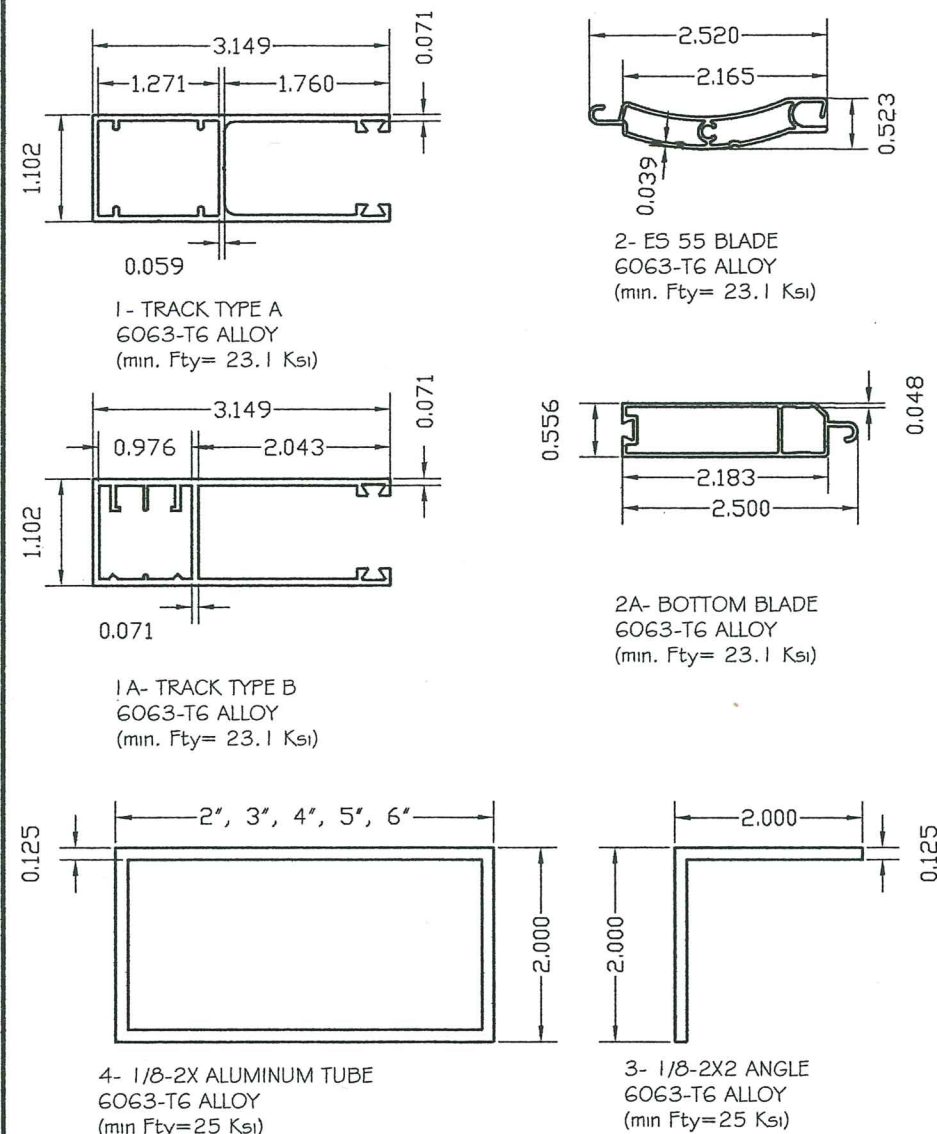
MIAMI-DADE BCCO:

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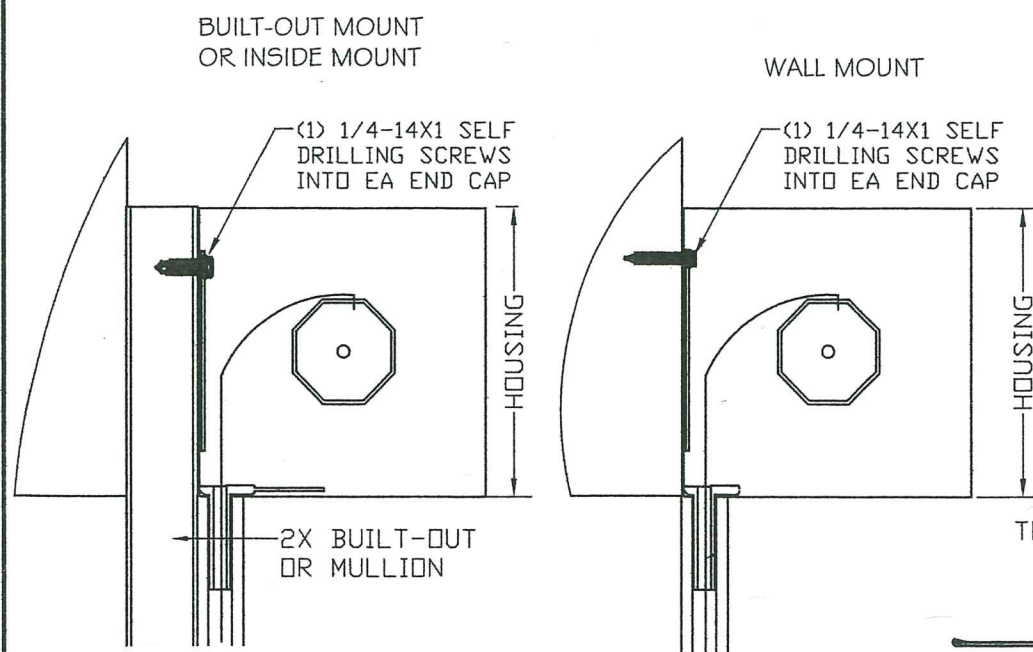
Manufacturer:
CROCI - North America
11600 ADELMO LANE
FORT MYERS, FL 33966
Tel.: (239) 278-3066 - Fax: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:	NA
DATE:	06/21/2020
DRAWING #	20-060
SHEET	1 OF 14

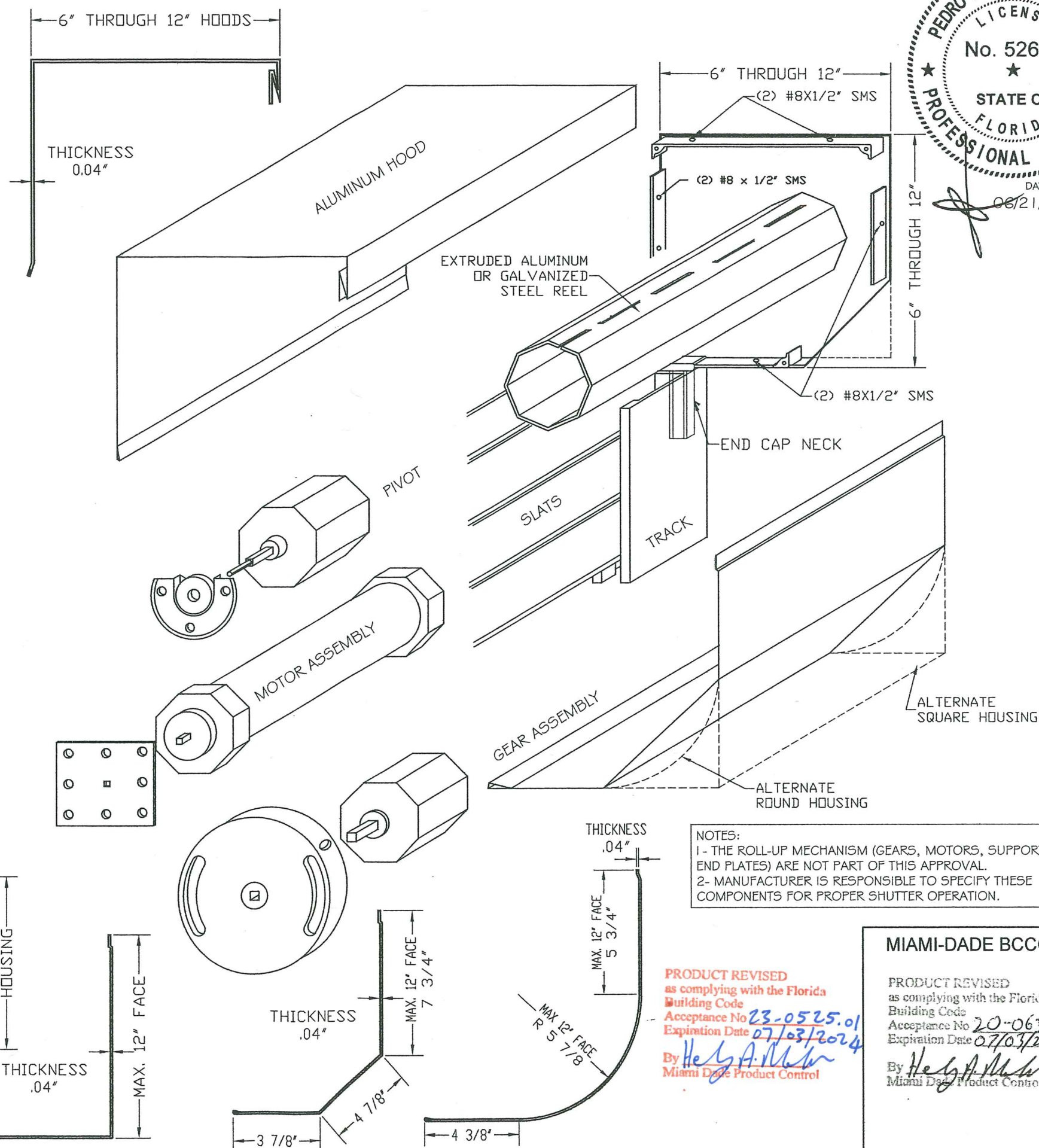
BASIC SHUTTER COMPONENTS



HOOD ANCHORAGE



HOOD ASSEMBLY



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By Healy A. Miller
Miami Dade Product Control

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By Healy A. Miller
Miami Dade Product Control

Manufacturer:

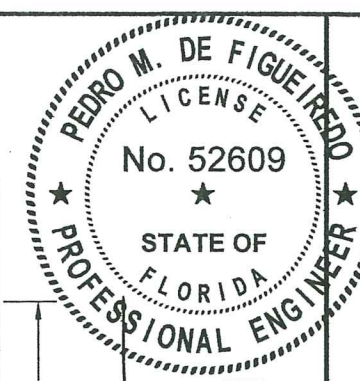
CROCI - North America
11600 ADELMO LANE
FORT MYERS, FL 33966
Tel.: (239) 278-3066 - Fax: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:
NA

DATE:
06/21/2020

DRAWING #
20-060

SHEET
2 OF 14



PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

Past & Present Moving the Future
ENGINEERING SERVICES - CA 32459
9715 W. BROWARD BLVD.
PLANTATION, FL 33324
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ENGINEER SEAL
PEDRO DE FIGUEIREDO
PE 52609

TABLE I - SLATS SPAN X DESIGN LOAD

POSITIVE & NEGATIVE LOADS (PSF)
X
MAXIMUM SLAT SPAN (IN)

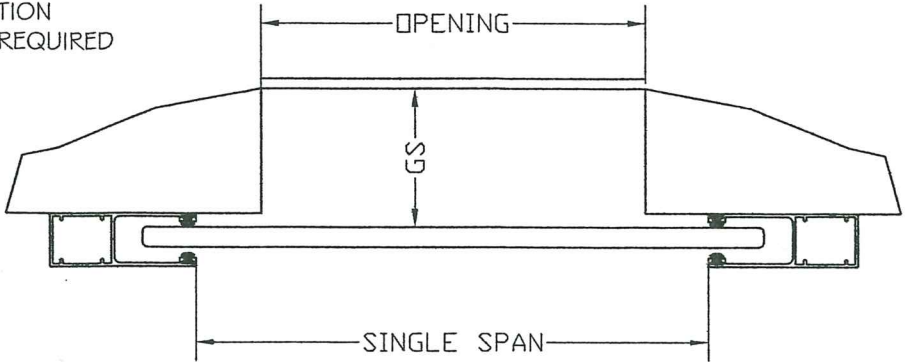
Pd PSF	1 SPAN	2 SPAN	3 SPAN
100	46	46	46
95	47	47	47
90	48	48	48
85	48	48	48
80	49	49	49
75	51	51	51
70	52	52	52
65	53	53	53
60	54	54	54
55	56	56	56
50	58	58	58
45	60	60	60
40	62	62	62
35	64	64	64

PD= DESIGN LOAD IN PSF
SPAN= SLAT SPAN IN INCHES

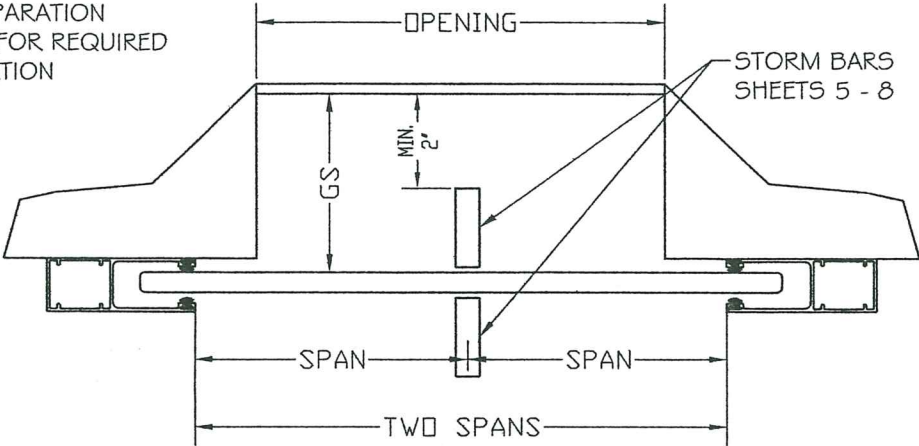
TABLE 1A - GLASS SEPARATION

SHUTTER SPAN	UNIT ELEVATION	
	<=30 FEET	>30 FEET
> 48"	3"	3"
<=48"	2 5/8"	2 5/8"

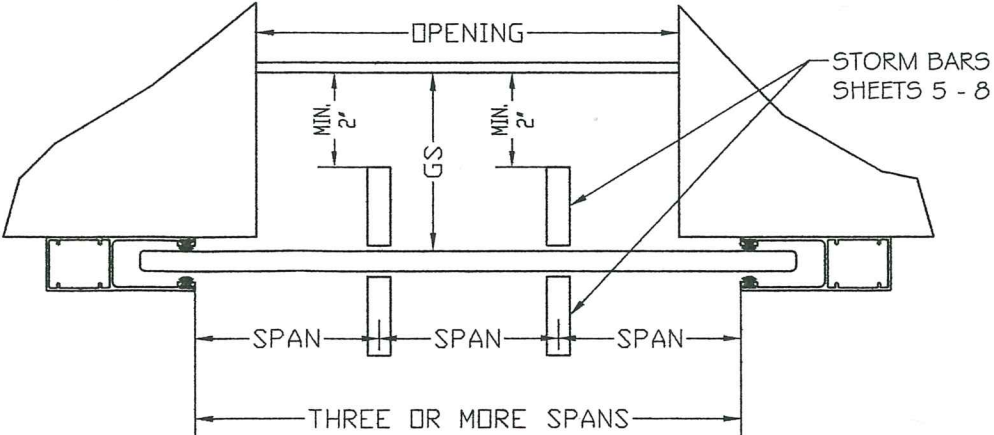
GS: GLASS SEPARATION
SEE TABLE 1A FOR REQUIRED
GLASS SEPARATION



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SEE TABLE 1A FOR REQUIRED
GLASS SEPARATION



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Miami Dade Product Control

MIAMI-DADE BCCO:

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TRACK ANCHORAGE OPTIONS

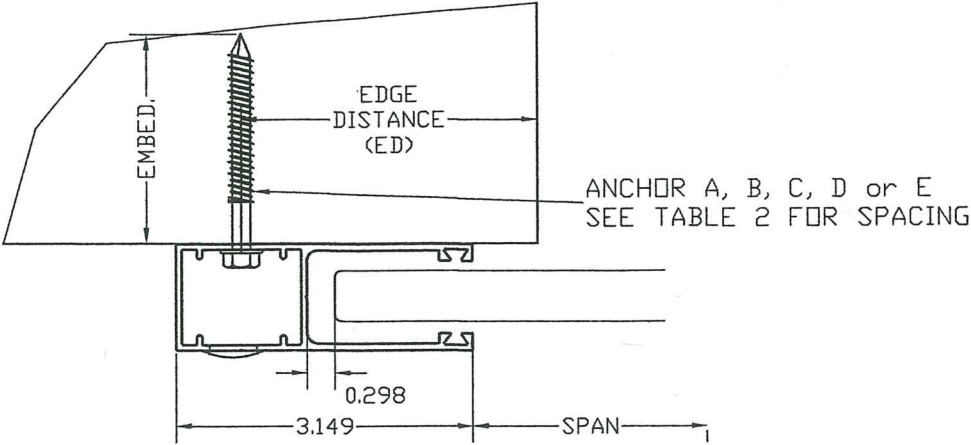


TABLE 2 - WALL MOUNT TRACK ANCHORAGE SPACING

Pd	SLAT SPAN (inches)									
	64in		58in		54in		51in		48in	
	A	B	C	D	E	A	B	C	D	E
35	6	3	6	6	6	6	4	6	6	6
50	6	3	6	6	6	6	4	6	6	6
60	-	-	-	-	-	6	3	6	6	6
75	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Pd= Design load in Psf // SPAN= Slat span in inches
Anchor spacing in inches on center

ANCHOR SCHEDULE

TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
A	1/4" TAPCON BY ITW	1 3/4"	3"	3000 psi CONCRETE
B	1/4" TAPCON BY ITW	1 1/4"	4"	HOLLOW BLOCK
C	5/16" TAPCON BY ITW	2 1/4"	4"	GF BLOCK
D	5/16" TAPCON BY ITW	1 3/4"	4"	HOLLOW BLOCK
E	1/4" TAPCON/ LAGS BY ITW	1 3/4"	1"	WOOD (SG>=.55)

NOTE: 1- ED - EDGE DISTANCE
2- EMBEDMENT IS BEYOND THE WALL DRESSING

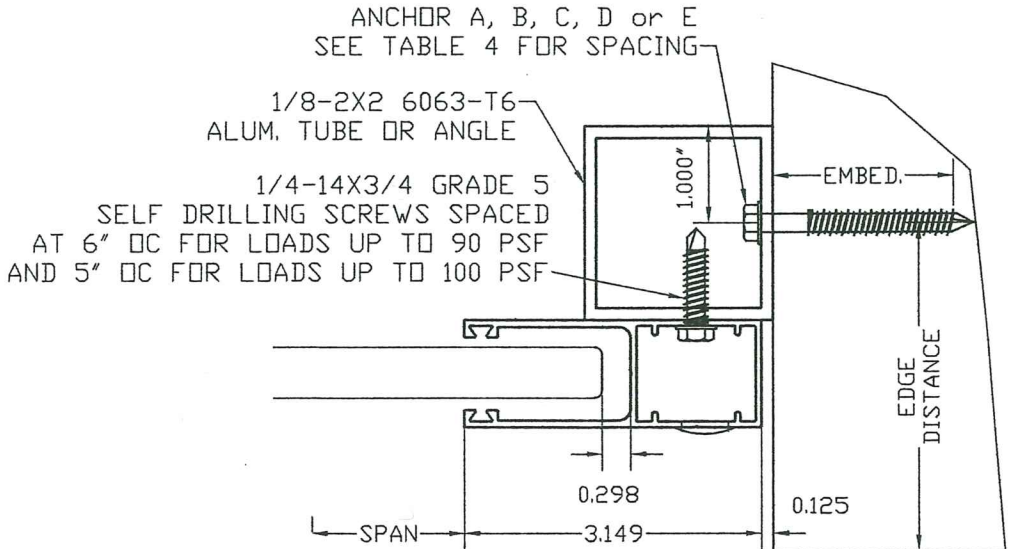


TABLE 3 - BUILT-OUT MOUNT TRACK ANCHORAGE SPACING

Pd	SLAT SPAN (inches)									
	64in		58in		54in		51in		48in	
	A	B	C	D	E	A	B	C	D	E
35	6	5	6	6	6	6	4	6	6	6
50	6	3	6	6	6	6	4	6	6	6
60	-	-	-	-	-	6	3	6	6	6
75	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Pd= Design load in Psf // SPAN= Slat span in inches
Anchor spacing in inches on center

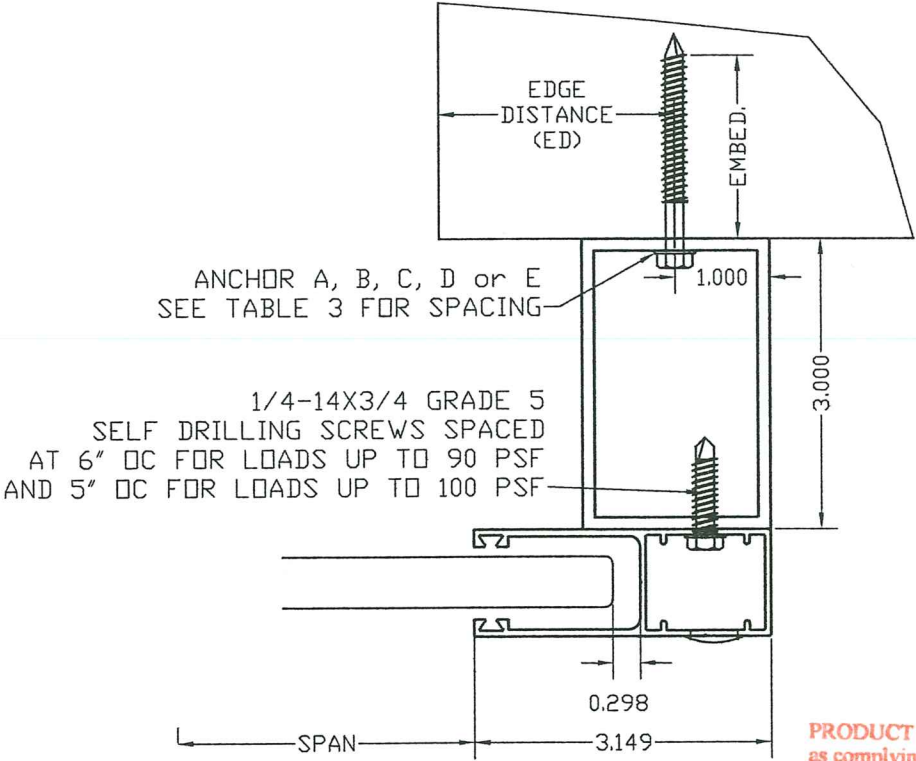


TABLE 4 - INSET MOUNT TRACK ANCHORAGE SPACING

Pd	SLAT SPAN (inches)									
	64in		58in		54in		51in		48in	
	A	B	C	D	E	A	B	C	D	E
35	6	4	6	6	6	6	4	6	6	6
50	6	-	6	6	6	6	3	6	6	6
60	-	-	-	-	-	6	5	6	6	6
75	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Pd= Design load in Psf // SPAN= Slat span in inches
Anchor spacing in inches on center



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PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

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SCALE:
NA
DATE:
06/21/2020
DRAWING #
20-060
SHEET
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STORM BARS SELECTION TABLES
ALLOWABLE DESIGN LOAD - Pd (Psf)

2x2 STORM BAR - 2 SPANS

SLAT	STORM BAR HEIGHT				
Span	36	48	60	72	84
46	100	71	46	31	--
48	90	69	44	30	--
51	75	65	41	--	--
54	60	60	39	--	--
58	50	50	36	--	--
64	35	35	33	--	--

2x2 STORM BAR - 3+ SPANS

SLAT	STORM BAR HEIGHT				
Span	36	48	60	72	84
46	100	81	52	35	--
48	90	78	50	35	--
51	75	74	46	--	--
54	60	60	45	--	--
58	50	50	41	--	--
64	35	35	35	--	--

2x3 STORM BAR - 2 SPANS

SLAT	STORM BAR HEIGHT				
Span	48	60	72	84	96
46	100	82	56	41	31
48	90	78	54	39	30
51	75	74	50	37	--
54	60	60	48	35	--
58	50	50	45	33	--
64	35	35	35	30	--

2x3 STORM BAR - 3+ SPANS

SLAT	STORM BAR HEIGHT				
Span	48	60	72	84	96
46	100	93	64	46	35
48	90	88	61	45	34
51	75	75	57	42	32
54	60	60	55	40	30
58	50	50	50	37	--
64	35	35	35	34	--

2x4 STORM BAR - 2 SPANS

SLAT	STORM BAR HEIGHT				
Span	72	84	96	108	120
46	86	62	47	37	--
48	82	60	46	35	--
51	75	56	42	34	--
54	60	53	40	31	--
58	50	50	38	30	--
64	35	35	34	--	--

2x4 STORM BAR - 3+ SPANS

SLAT	STORM BAR HEIGHT				
Span	72	84	96	108	120
46	97	71	54	42	31
48	90	68	52	40	30
51	75	64	48	38	--
54	60	60	45	35	--
58	50	50	43	34	--
64	35	35	35	30	--

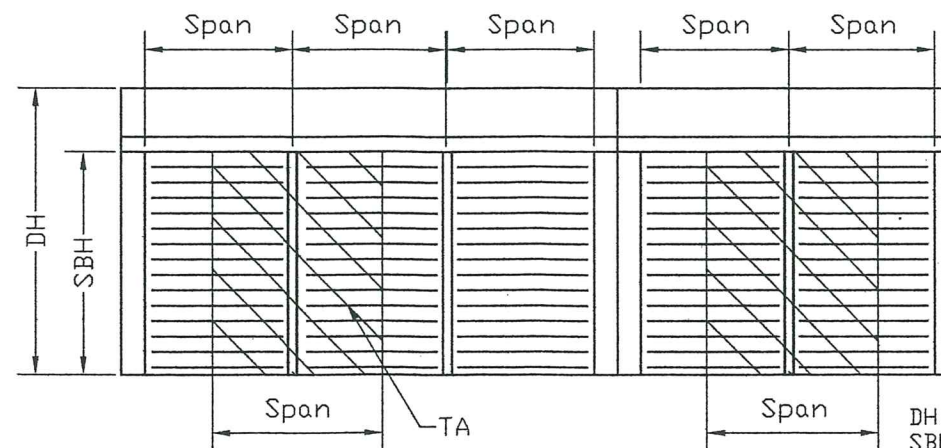
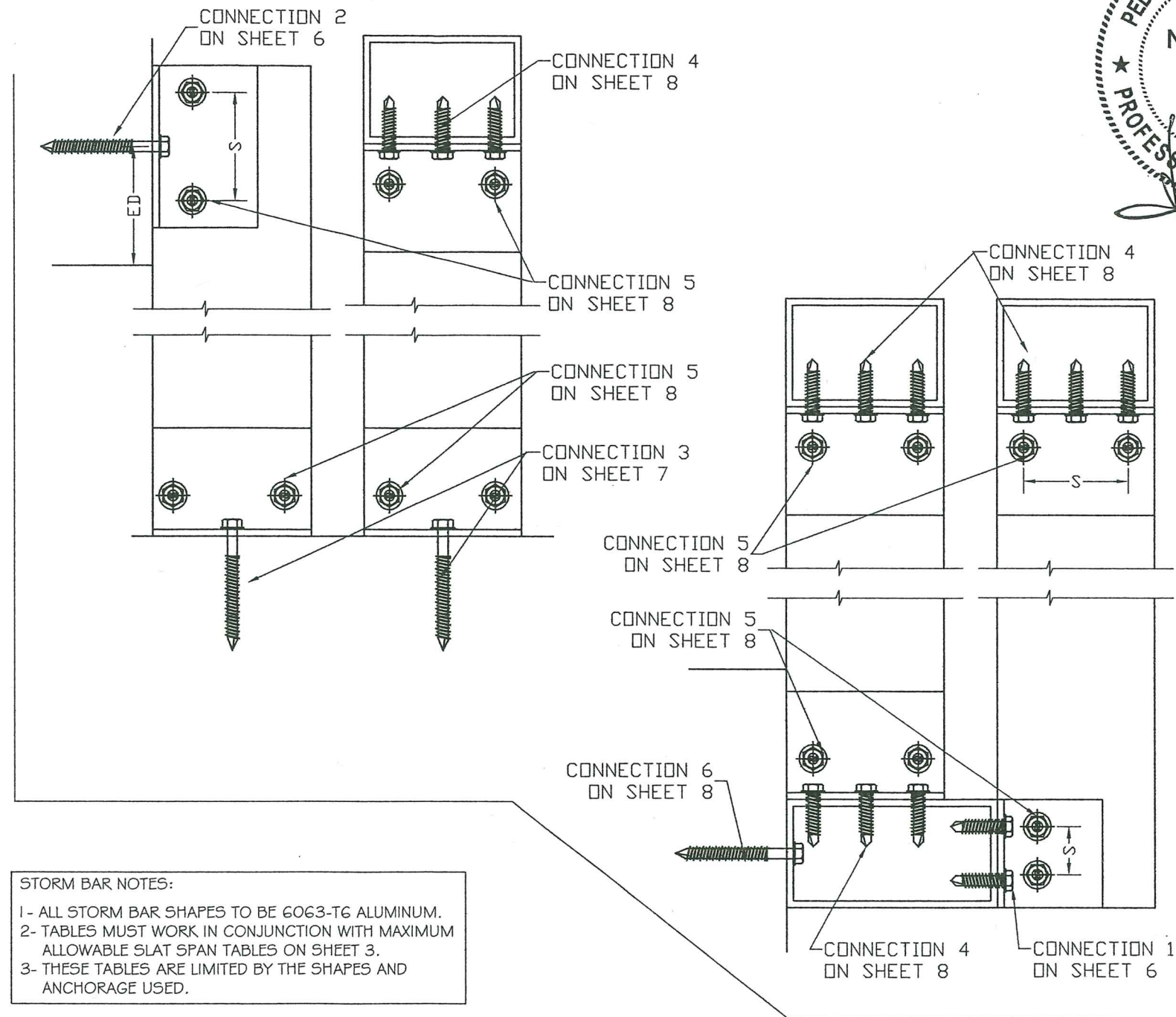
2x5 STORM BAR - 2 SPANS

SLAT	STORM BAR HEIGHT				
Span	72	84	96	108	120
46	100	87	66	52	42
48	90	83	63	50	40
51	75	75	59	46	38
54	60	60	56	44	35
58	50	50	50	41	33
64	35	35	35	35	30

2x5 STORM BAR - 3+ SPANS

SLAT	STORM BAR HEIGHT				
Span	72	84	96	108	120
46	100	99	75	59	47
48	90	90	72	56	45
51	75	75	67	53	43
54	60	60	60	50	40
58	50	50	50	46	37
64	35	35	35	35	34

STORM BARS MOUNTING OPTIONS

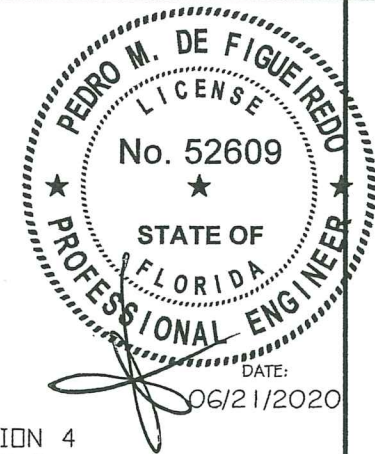


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By *Hegya M*
Miami Dade Product Control

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DH - DOOR HEIGHT
SBH - STORM BAR HEIGHT
TA - STORM BAR TRIBUTARY AREA



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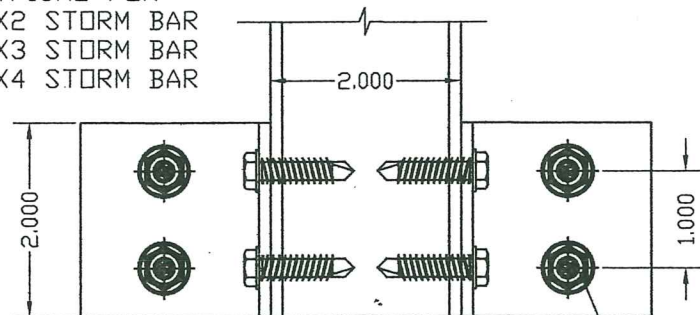
PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
CROCI - North America
11600 ADELMO LANE
FORT MYERS, FL 33966
Tel.: (239) 278-3066 - Fax: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

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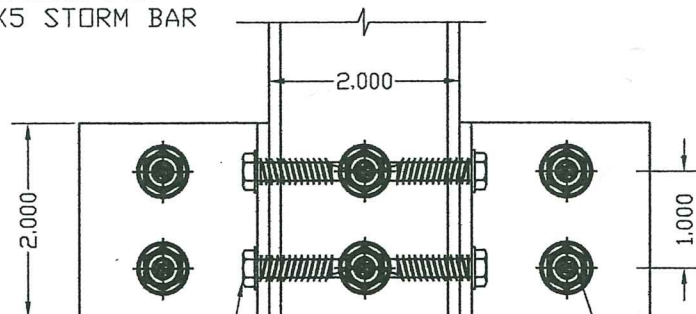
CONNECTION 1 STORM BARS ANCHORAGE WALL MOUNTED TO ALUMINUM TUBE

TYPICAL FOR
2X2 STORM BAR
2X3 STORM BAR
2X4 STORM BAR



2 PER EA CLIP LEG
1/4-14X3/4" SELF
DRILLING SCREW

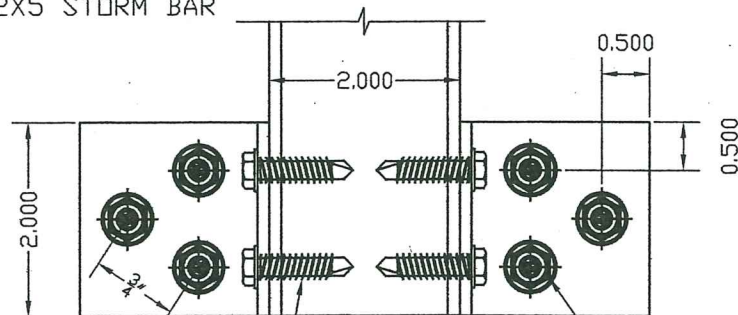
TYPICAL FOR
2X5 STORM BAR



CONNECTION 5

6 TOTAL
1/4-14X3/4" SELF
DRILLING SCREW

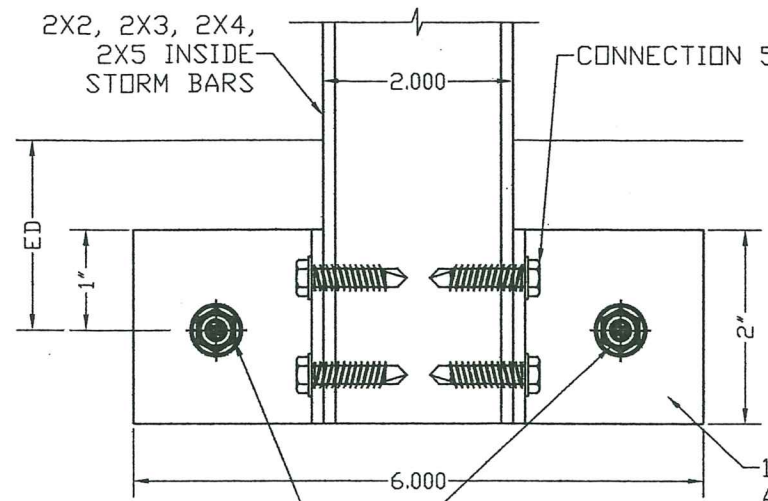
OPTIONAL CONNECTION
FOR 2X5 STORM BAR



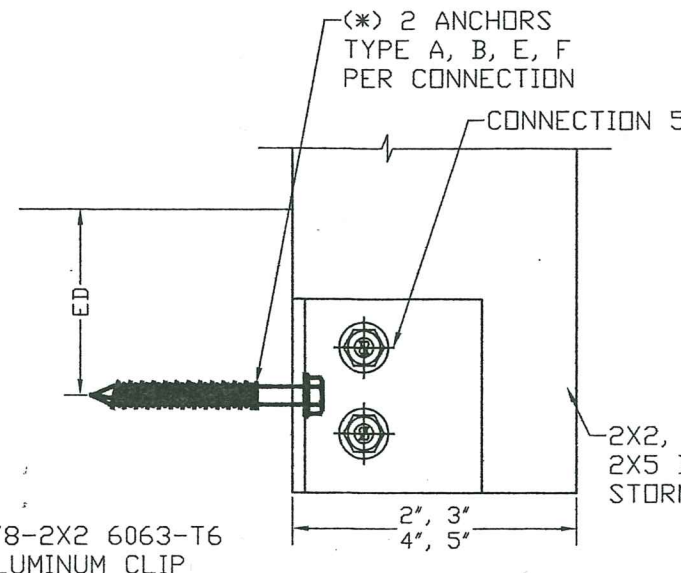
CONNECTION 5

6 TOTAL
1/4-14X3/4" SELF
DRILLING SCREW

CONNECTION 2 STORM BARS ANCHORAGE TO WALL



(*) 2 ANCHORS
TYPE A, B, E, F
PER CONNECTION



(*) 2 ANCHORS
TYPE A, B, E, F
PER CONNECTION

CONNECTION 5

2X2, 2X3, 2X4,
2X5 INSIDE
STORM BARS

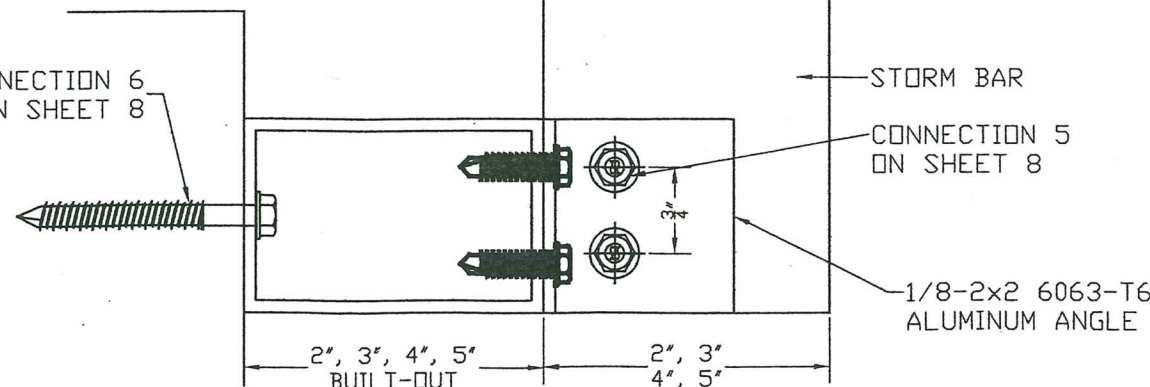
(*) CONDITION IS ONLY APPLICABLE
FOR POSITIVE LOAD ON INSIDE STORM BARS
LOAD TRANSFERED BY BEARING TO SUPPORTING STRUCTURE

ANCHOR SCHEDULE

TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
A	1/4" TAPCON BY ITW	1 3/4"	3"	3000 psi CONCRETE
B	1/4" TAPCON BY ITW	1 1/4"	3"	HOLLOW BLOCK
E	1/4" TAPCON/ LAGS BY ITW	1 3/4"	1"	WOOD (SG >= .55)
F	1/4"-20 SOLID SET	7/8"	3"	3000 psi CONCRETE

NOTE: 1- ED - EDGE DISTANCE
2- EMBEDMENT IS BEYOND THE WALL DRESSING

CONNECTION 6
ON SHEET 8



STORM BAR

CONNECTION 5
ON SHEET 8

1/8-2x2 6063-T6
ALUMINUM ANGLE

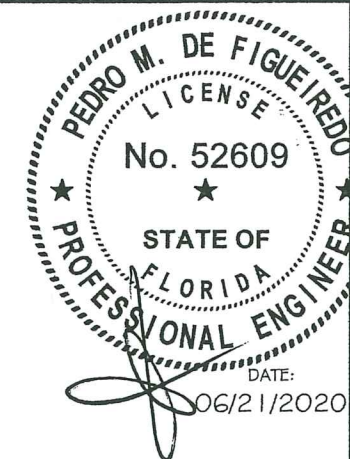
2", 3", 4", 5"
BUILT-OUT

2", 3"
4", 5"

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By *Helga A. Mahr*
Miami Dade Product Control

MIAMI-DADE BCCO:

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By *Helga A. Mahr*
Miami Dade Product Control



PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
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EMAIL: INFO@CROCIUSA.COM

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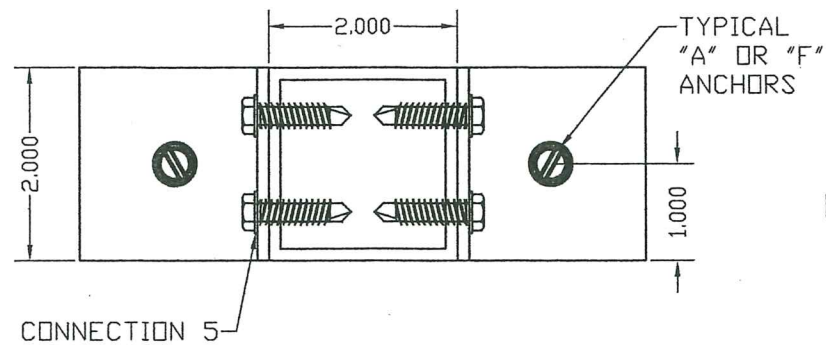
DATE:
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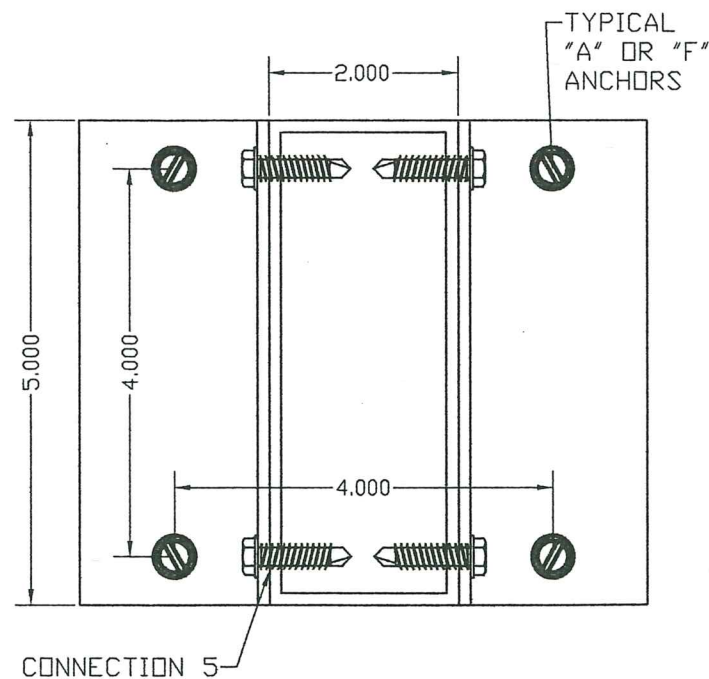
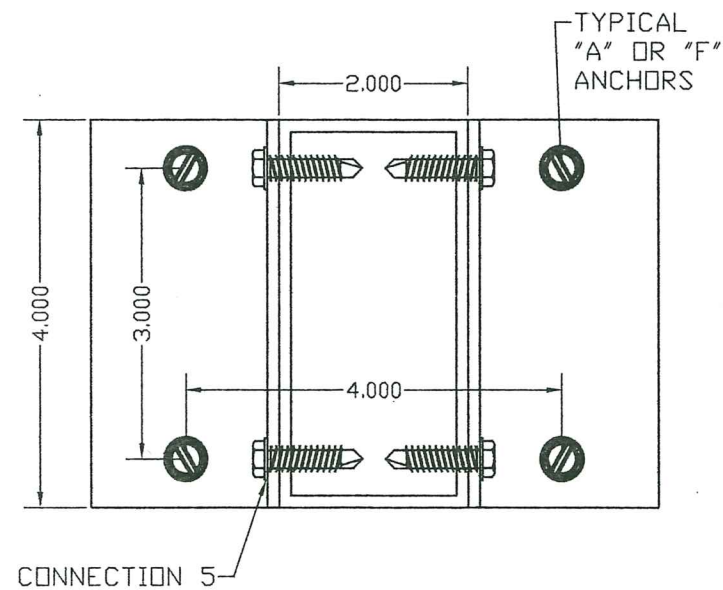
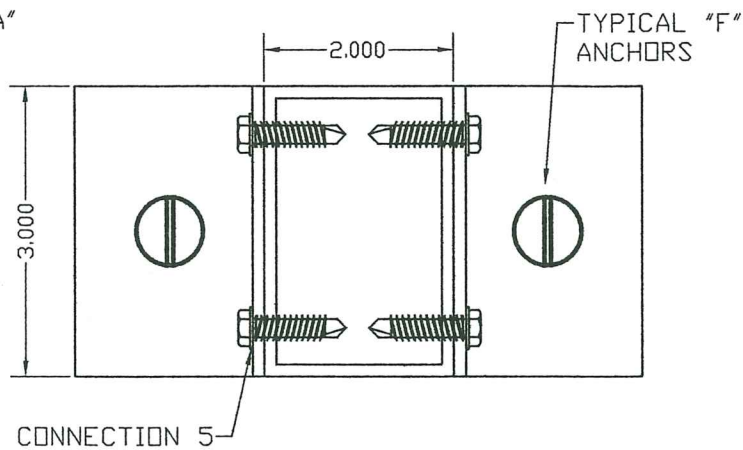
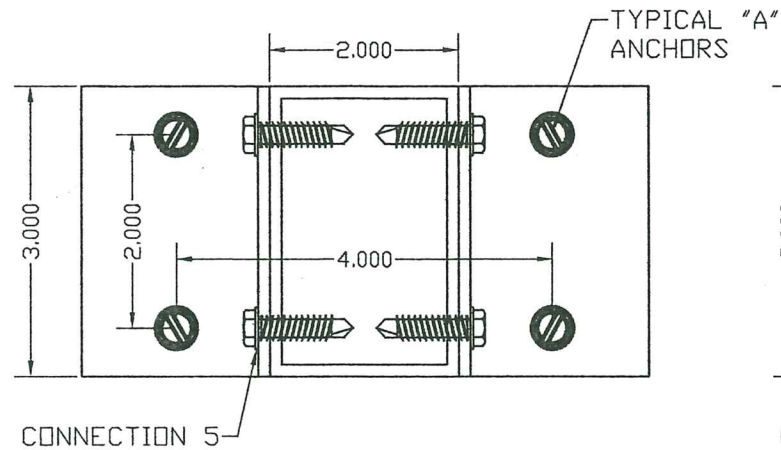
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CONNECTION 3
STORM BARS ANCHORAGE
FLOOR/CEILING MOUNTED TO CONCRETE

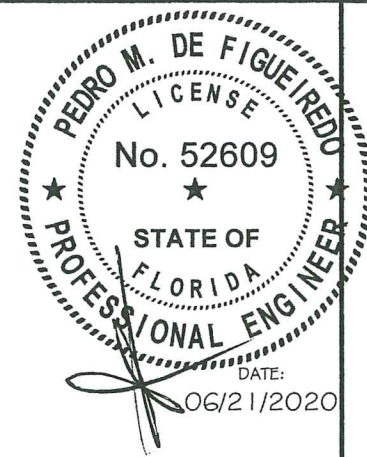
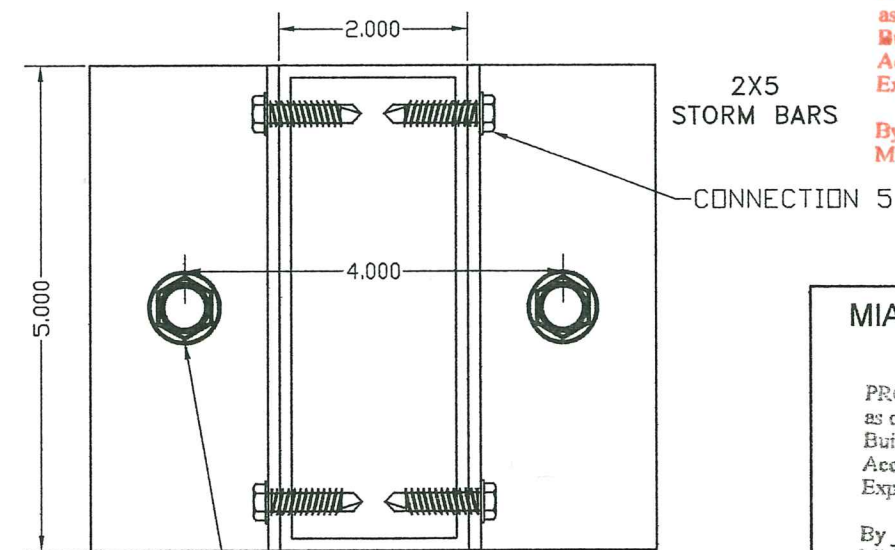
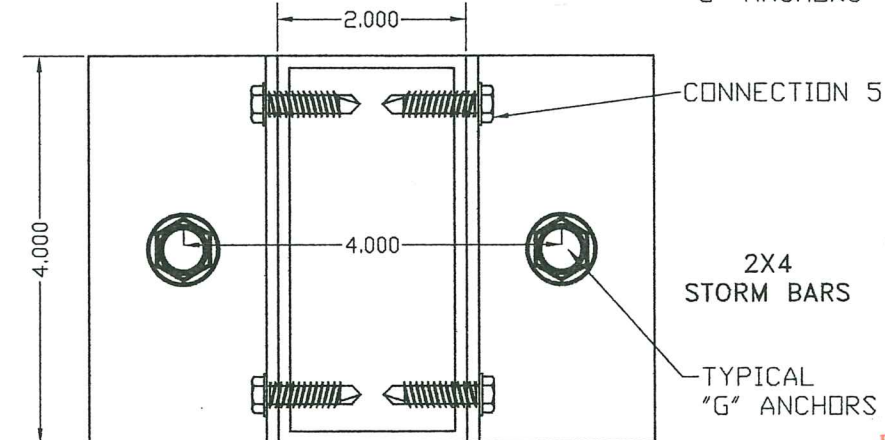
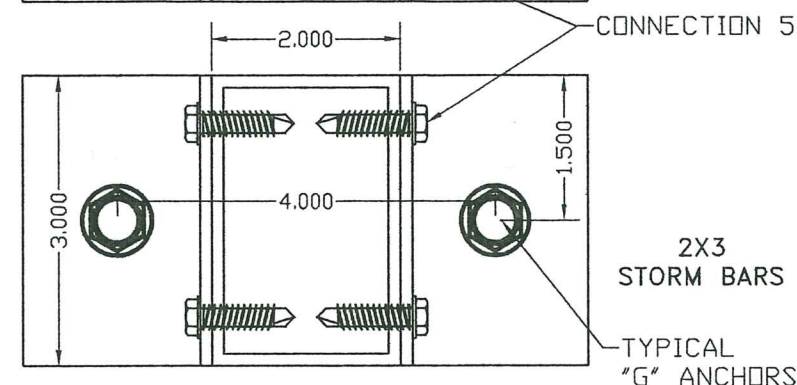
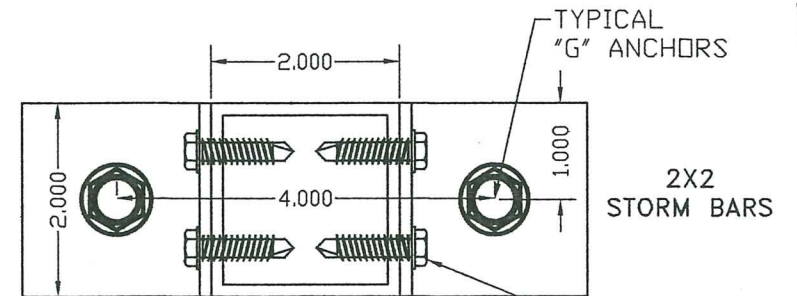


ANCHOR SCHEDULE

TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
A	1/4" TAPCON BY ITW	1 3/4"	3"	3000 pSI CONCRETE
F	1/4"-20 SOLID SET	7/8"	3"	3000 pSI CONCRETE
G	3/8" WEDGE BOLTS BY POWERS	3"	4"	3000 pSI CONCRETE

NOTE: ED - EDGE DISTANCE

TYPE "G" ANCHORS APPLICABLE FOR ALL CASES
MUST NOT EXCEED STORM BAR DESIGNED LOAD



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ES 55 SHUTTER
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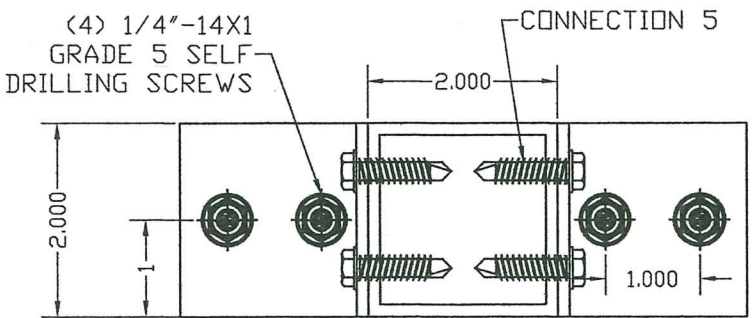
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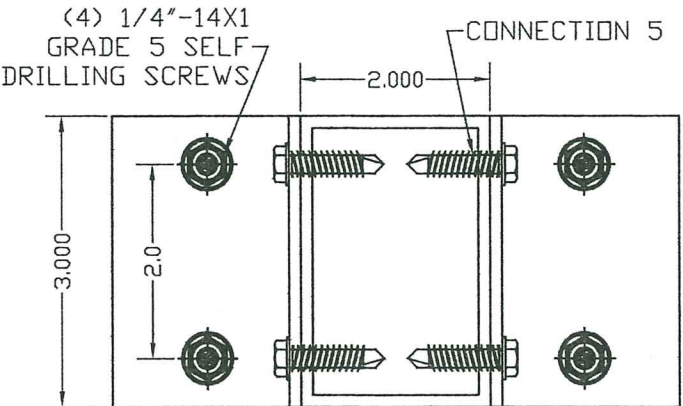
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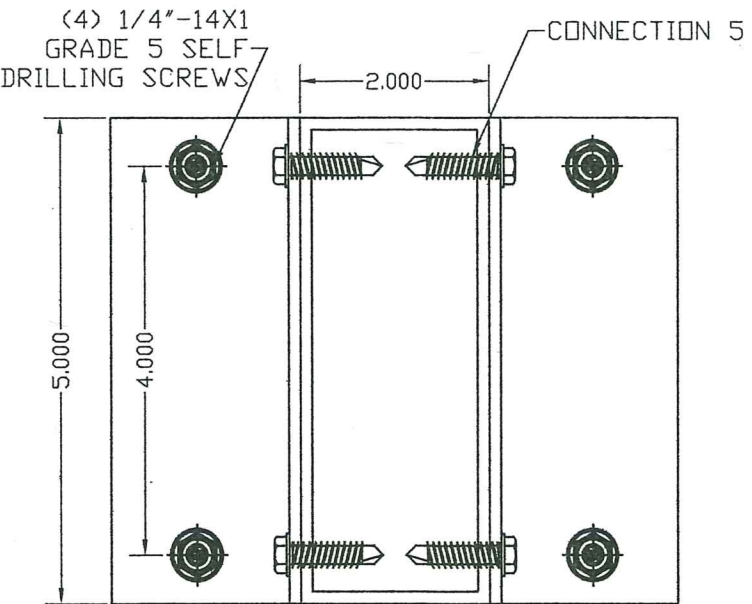
CONNECTION 4
STORM BARS ANCHORAGE FLOOR/CEILING
TO BUILT-OUT TUBE OR HEADER



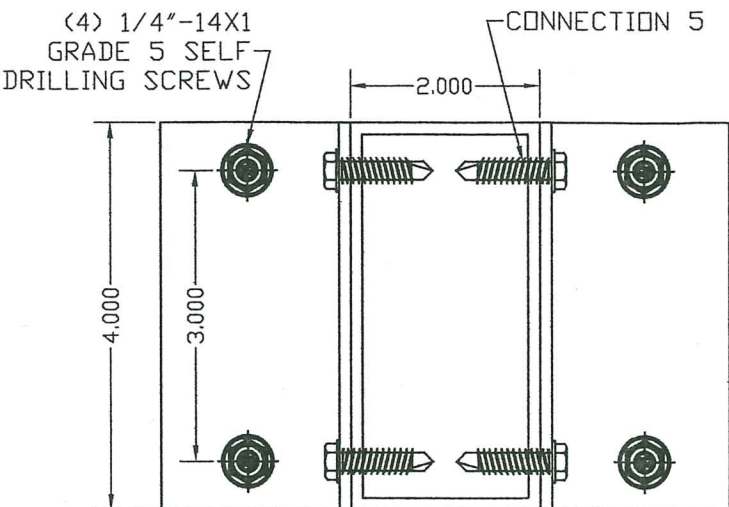
2X2 STORM BARS
ACCEPTABLE FOR ALL CASES
MUST NOT EXCEED STORM BAR DESIGNED LOAD



2X3 STORM BARS
ACCEPTABLE FOR ALL CASES
MUST NOT EXCEED STORM BAR DESIGNED LOAD



2X5 STORM BARS
ACCEPTABLE FOR ALL CASES
MUST NOT EXCEED STORM BAR DESIGNED LOAD



2X4 STORM BARS
ACCEPTABLE FOR ALL CASES
MUST NOT EXCEED STORM BAR DESIGNED LOAD

CONNECTION 6
MINIMUM NUMBER OF ANCHORS
TO SECURE BUILT-OUT TO WALL

STORM BARS	ANCHOR TYPE			
	A	B	E	D
2x2	2	8	2	4
2x3	2	10	2	5
2x4	3	-	3	6
2x5	3	-	3	7

S	ANCHOR TYPE			
	A	B	E	D
S	4.0	4.0	2.0	3.75

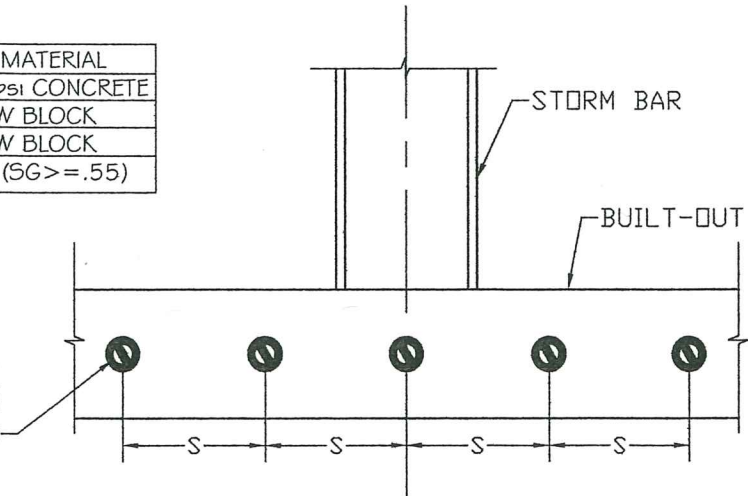
S - ANCHOR SPACING

ANCHOR SCHEDULE

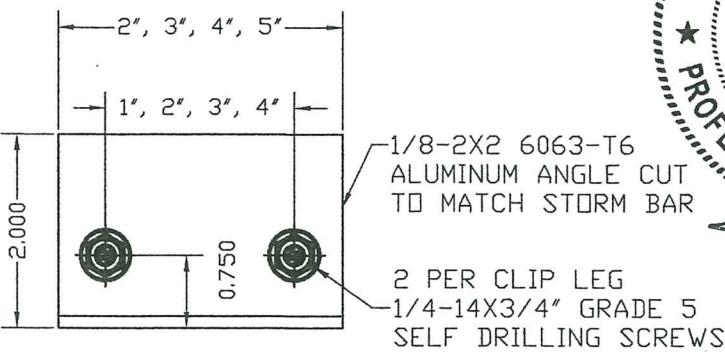
TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
A	1/4" TAPCON BY ITW	1 3/4"	3"	3000 psi CONCRETE
B	1/4" TAPCON BY ITW	1 1/4"	4"	HOLLOW BLOCK
D	5/16" TAPCON BY ITW	1 3/4"	4"	HOLLOW BLOCK
E	1/4" TAPCON/ LAGS BY ITW	1 3/4"	1"	WOOD (SG >= .55)

NOTE: ED - EDGE DISTANCE

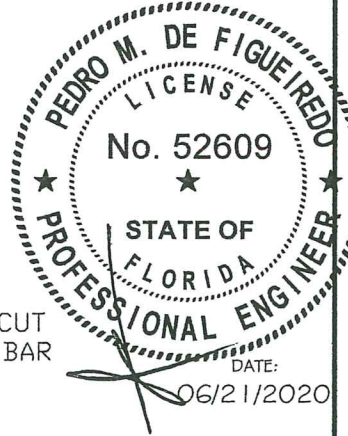
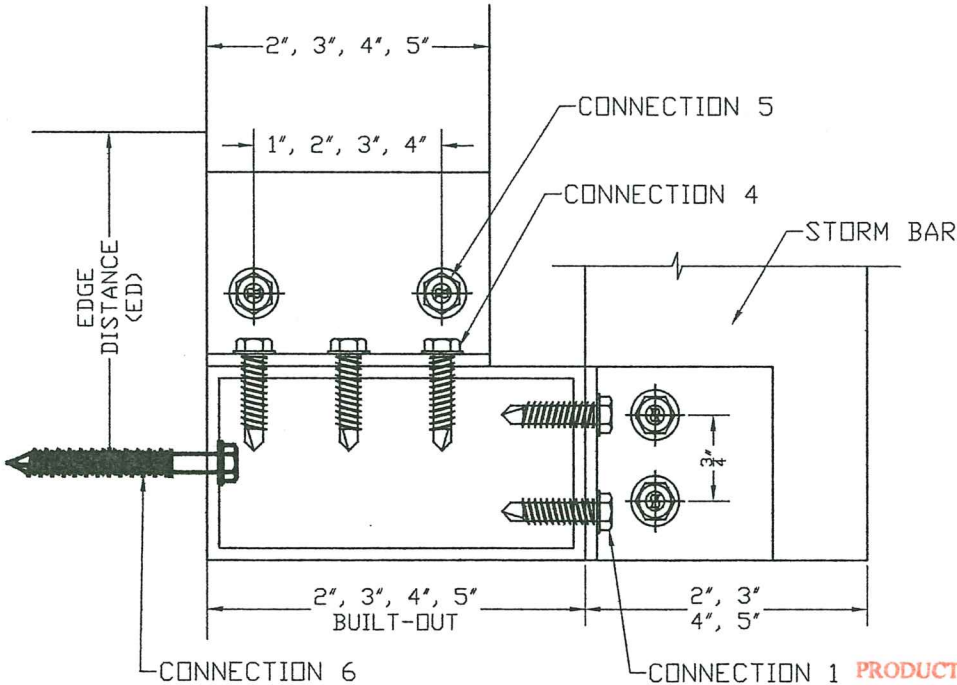
QUANTITY OF ANCHORS FROM TABLE
ARE TO BE EQUALLY DIVIDED
FROM CENTER LINE OF STORM BAR



CONNECTION 5
CLIP ANCHORED TO STORM BAR



CONNECTION 6
BUILT-OUT WALL ANCHORAGE



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PE 52609



PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
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EMAIL: INFO@CROCIUSA.COM

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By H. G. A. M. B.
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HEADER SELECTION TABLES ALLOWABLE DESIGN LOAD - Pd (Psf)

MAXIMUM DESIGN PRESSURE
FOR 2X3 HEADER - 2 SPANS

SLAT	STORM BAR HEIGHT					
TT	36	48	60	72	84	96
92	88	66	53	44	38	33
96	81	61	49	40	35	30
102	72	54	43	36	31	--
108	64	48	38	32	--	--
116	55	41	33	--	--	--
128	45	34	--	--	--	--

MAXIMUM DESIGN PRESSURE
FOR 2X3 HEADER - 3+ SPANS

SLAT	STORM BAR HEIGHT					
TT	36	48	60	72	84	96
108	82	61	49	41	35	30
120	66	50	40	33	--	--
126	60	45	36	30	--	--
138	49	37	--	--	--	--
144	41	31	--	--	--	--
153	32	--	--	--	--	--

MAXIMUM DESIGN PRESSURE
FOR 2X4 HEADER - 2 SPANS

SLAT	STORM BAR HEIGHT					
TT	48	60	72	84	96	108
92	100	81	67	57	50	45
96	90	74	62	53	46	41
102	75	65	54	47	41	36
108	60	58	49	42	36	32
116	50	50	42	36	31	--
128	35	35	34	--	--	--

MAXIMUM DESIGN PRESSURE
FOR 2X4 HEADER - 3+ SPANS

SLAT	STORM BAR HEIGHT					
TT	48	60	72	84	96	108
108	94	75	62	53	47	41
120	76	60	50	43	38	33
138	57	46	38	32	--	--
144	52	42	35	30	--	--
153	46	37	31	--	--	--
162	39	31	--	--	--	--

MAXIMUM DESIGN PRESSURE
FOR 2X5 HEADER - 2 SPANS

SLAT	STORM BAR HEIGHT					
TT	60	72	84	96	108	120
92	100	94	80	70	62	56
96	90	86	74	65	57	52
102	75	75	65	57	51	46
108	60	60	58	51	45	41
116	50	50	50	44	39	35
128	35	35	35	35	32	--

MAXIMUM DESIGN PRESSURE
FOR 2X5 HEADER - 3+ SPANS

SLAT	STORM BAR HEIGHT					
TT	60	72	84	96	108	120
120	85	70	60	53	47	42
138	64	53	45	40	35	32
144	59	49	42	36	32	--
153	52	43	37	32	--	--
162	46	38	33	--	--	--
174	40	33	--	--	--	--

MAXIMUM DESIGN PRESSURE
FOR 2X6 HEADER - 2 SPANS

SLAT	STORM BAR HEIGHT					
TT	60	72	84	96	108	120
92	100	100	100	93	83	75
96	90	90	90	86	76	69
102	75	75	75	75	67	61
108	60	60	60	60	60	54
116	50	50	50	50	50	47
128	35	35	35	35	35	35

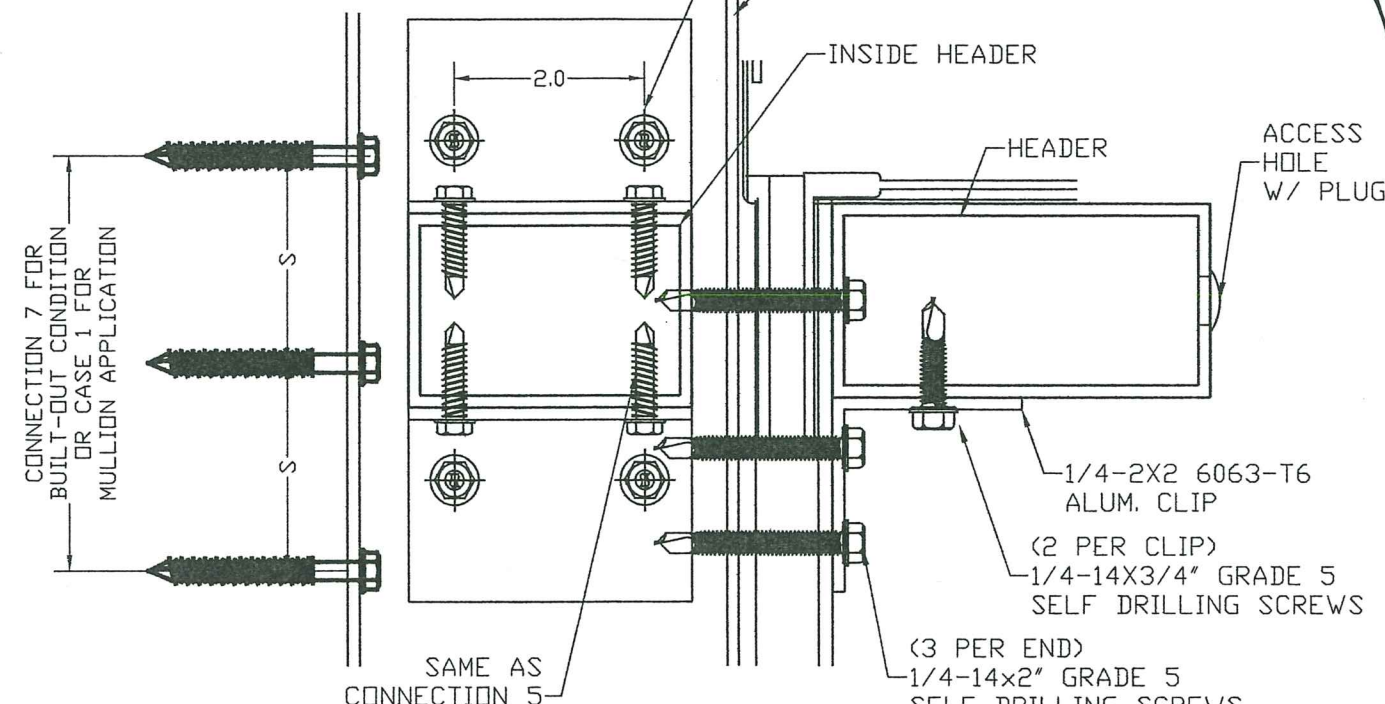
MAXIMUM DESIGN PRESSURE
FOR 2X6 HEADER - 3+ SPANS

SLAT	STORM BAR HEIGHT					
TT	60	72	84	96	108	120
138	85	71	60	53	47	42
144	78	65	56	49	43	39
153	69	57	49	43	38	34
162	60	51	44	38	34	30
174	50	44	38	33	--	--
192	35	35	31	--	--	--

CONNECTION 7
MINIMUM NUMBER OF ANCHORS
TO SECURE BUILT-OUT TO WALL
AT HEADER CONNECTION

		ANCHOR TYPE			
		A	B	E	G
n	2	10	2	2	
S	4.0	4.0	3.0	4.5	

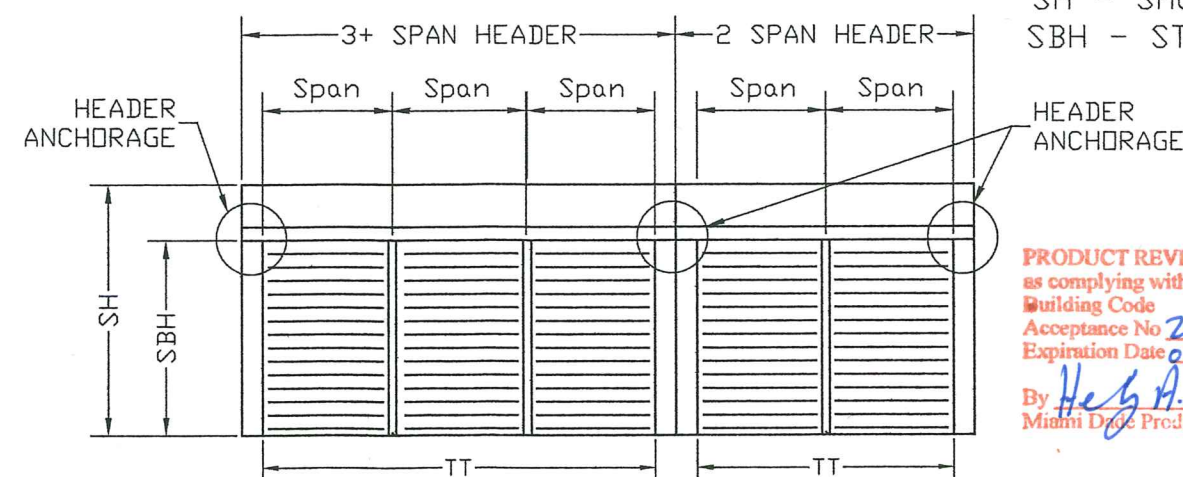
n - MINIMUM NUMBER OF ANCHORS
S - ANCHOR SPACING
ED - EDGE DISTNCE



ANCHOR SCHEDULE

TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
A	1/4" TAPCON BY ITW	1 3/4"	3"	3000 psi CONCRETE
B	1/4" TAPCON BY ITW	1 1/4"	4"	HOLLOW BLOCK
E	1/4" TAPCON/ LAGS BY ITW	1 3/4"	1"	WOOD (SG>=.55)
G	3/8" WEDGE BOLTS BY POWERS	2 1/2"	4"	HOLLOW BLOCK

NOTE: ED - EDGE DISTNCE

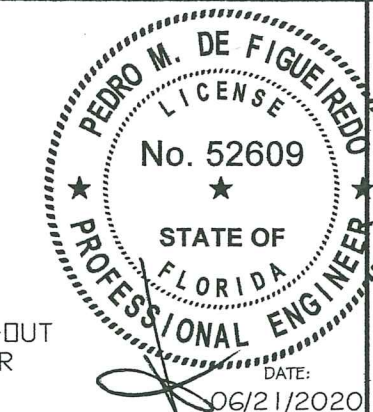


TT - TRACK TO TRACK
SH - SHUTTER HEIGHT
SBH - STORM BAR HEIGHT

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PRODUCT CATEGORY: SHUTTERS

PRODUCT MATERIAL: ALUMINUM

ES 55 SHUTTER

EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
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FORT MYERS, FL 33966

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EMAIL: INFO@CROCIUSA.COM

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MULLION SELECTION TABLES

ALLOWABLE DESIGN LOAD - Pd (Psf)

2x2 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	68	39	--	--	--	--
48	90	65	38	--	--	--	--
51	75	61	36	--	--	--	--
54	60	58	34	--	--	--	--
58	50	50	31	--	--	--	--
64	35	35	--	--	--	--	--

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x3 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	100	100	66	44	31	--
48	90	90	90	63	43	30	--
51	75	75	75	60	40	--	--
54	60	60	60	56	38	--	--
58	50	50	50	50	35	--	--
64	35	35	35	35	32	--	--

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x5 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	100	100	100	100	100	80
48	90	90	90	90	90	90	77
51	75	75	75	75	75	75	73
54	60	60	60	60	60	60	60
58	50	50	50	50	50	50	50
64	35	35	35	35	35	35	35

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x4 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	100	100	100	90	63	46
48	90	90	90	90	86	61	44
51	75	75	75	75	75	57	42
54	60	60	60	60	60	54	39
58	50	50	50	50	50	50	37
64	35	35	35	35	35	35	33

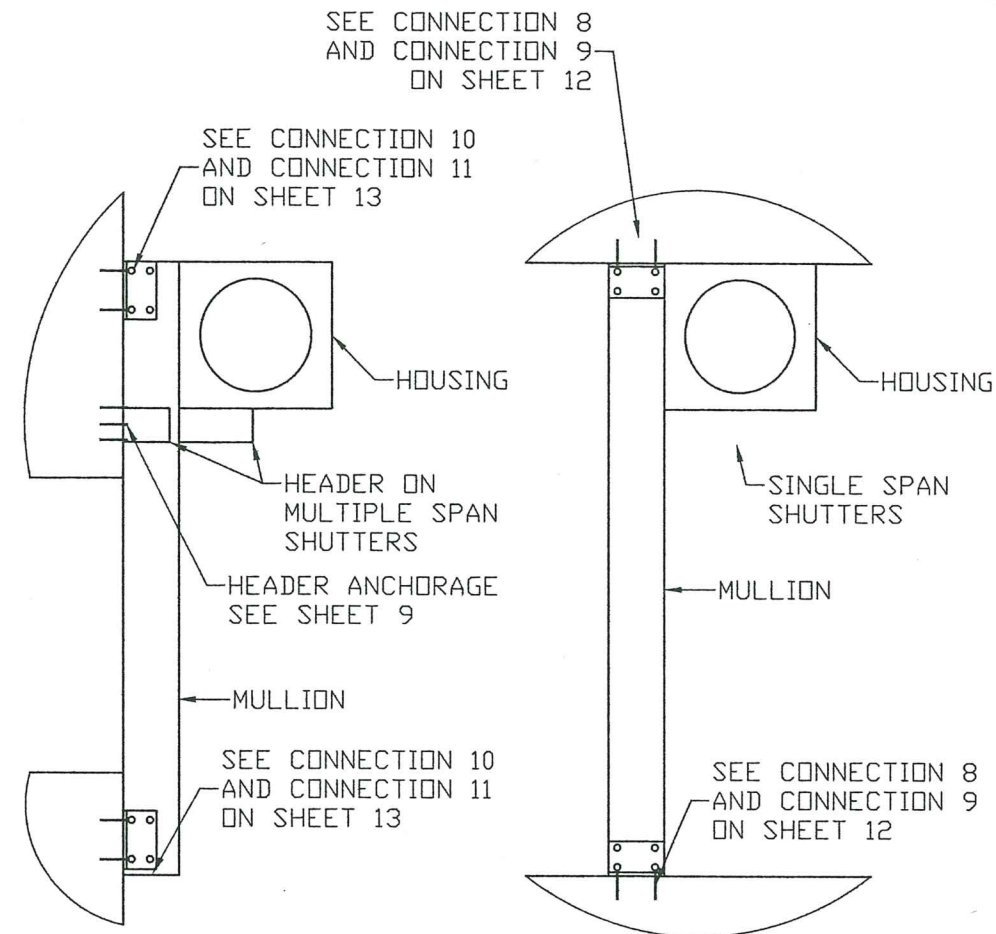
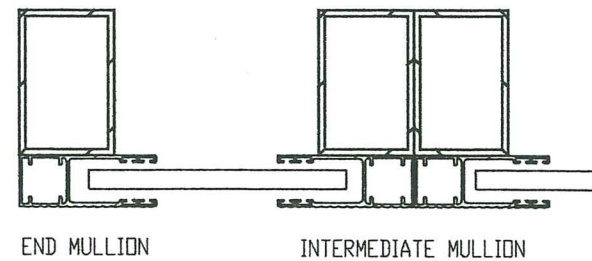
LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x6 MULLION

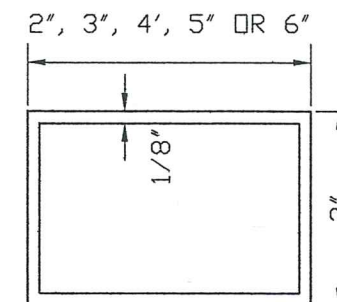
SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	100	100	100	100	100	100
48	90	90	90	90	90	90	90
51	75	75	75	75	75	75	75
54	60	60	60	60	60	60	60
58	50	50	50	50	50	50	50
64	35	35	35	35	35	35	35

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

CASE I : SINGLE SPAN SHUTTERS AND MULTIPLE SPAN SHUTTERS WITH HEADER ANCHORED TO WALL



SHAPES:
1/8-2X2 6063-T6
1/8-2X3 6063-T6
1/8-2X4 6063-T6
1/8-2X5 6063-T6
1/8-2X6 6063-T6



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as complying with the Florida
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Expiration Date 07/03/2024
By: *Heidi A. Miller*
Miami Dade Product Control

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as complying with the Florida
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Acceptance No 20-0670.17
Expiration Date 07/01/23
By: *Heidi A. Miller*
Miami Dade Product Control

PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
CROCI -North America
11600 ADELMO LANE
FORT MYERS, FL 33966
Tel.: (239) 278-3066 - Fax: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:
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ENGINEER SEAL
PEDRO DE FIGUEIREDO
PE 52609

MULLION SELECTION TABLES
ALLOWABLE DESIGN LOAD - Pd (Psf)

2x2 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	67	40	--	--	--	--	--
48	64	38	--	--	--	--	--
51	60	36	--	--	--	--	--
54	57	34	--	--	--	--	--
58	50	32	--	--	--	--	--
64	35	--	--	--	--	--	--

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x3 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	89	67	45	32	--	--
48	90	85	64	43	31	--	--
51	75	75	60	41	--	--	--
54	60	60	57	38	--	--	--
58	50	50	50	36	--	--	--
64	35	35	35	32	--	--	--

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x5 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	89	74	64	56	50	45
48	90	85	71	61	54	48	43
51	75	75	67	58	51	45	41
54	60	60	60	55	48	43	38
58	50	50	50	50	44	40	36
64	35	35	35	35	35	35	32

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x4 MULLION

SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	89	74	64	56	47	36
48	90	85	71	61	54	45	35
51	75	75	67	58	51	42	32
54	60	60	60	55	48	40	31
58	50	50	50	50	44	37	--
64	35	35	35	35	35	34	--

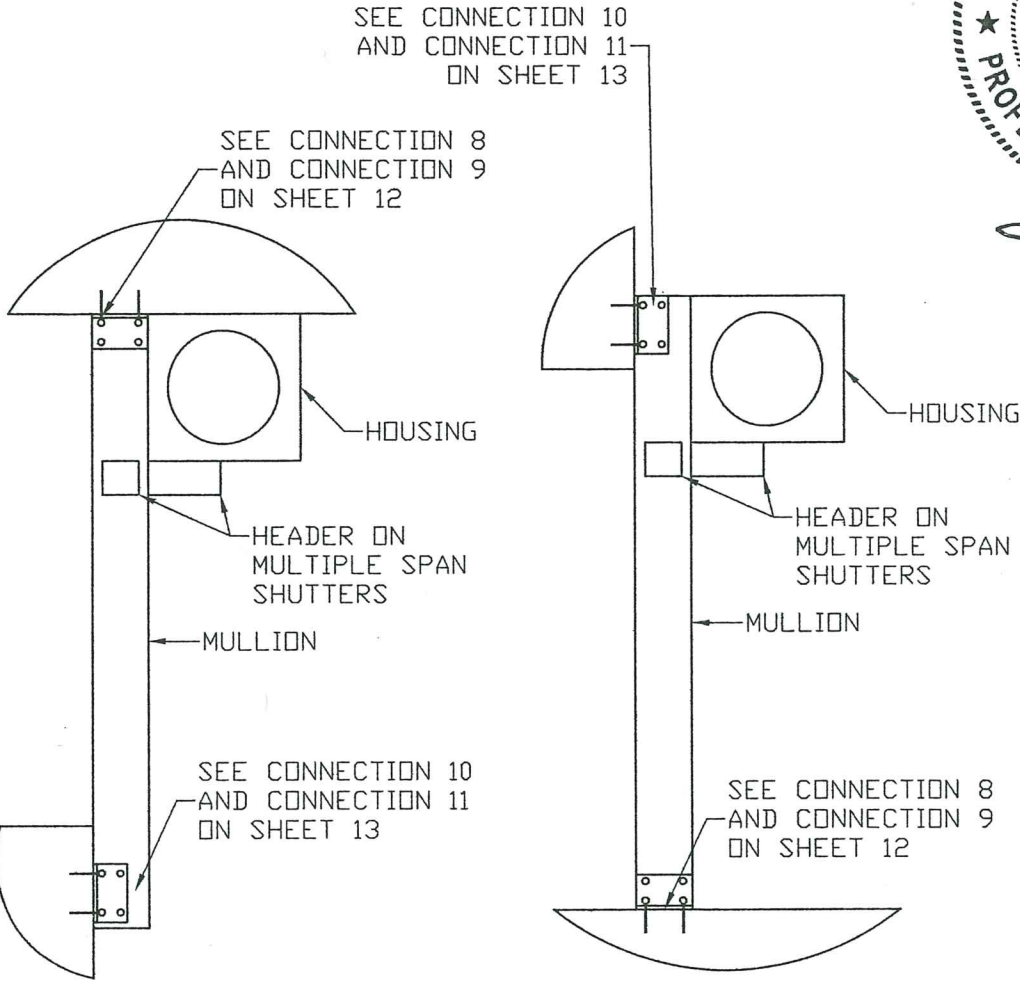
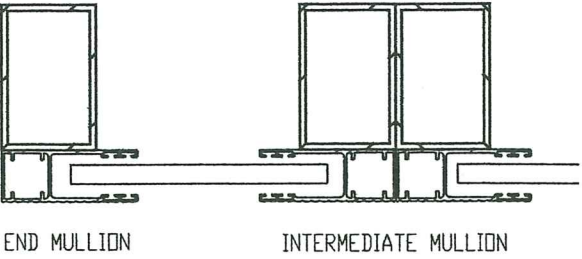
LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

2x6 MULLION

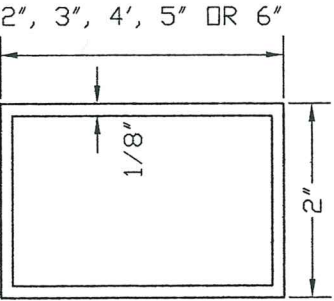
SLAT SPAN	MULLION HEIGHT						
	48	60	72	84	96	108	120
46	100	89	74	64	56	50	45
48	90	85	71	61	54	48	43
51	75	75	67	58	51	45	41
54	60	60	60	55	48	43	38
58	50	50	50	50	44	40	36
64	35	35	35	35	35	35	32

LOADS ARE LIMITED BY THE SLAT SPAN TABLE ON SHEET 3

CASE 2:
MULTIPLE SPAN SHUTTERS WITH
HEADER ANCHORED TO MULLION



SHAPES:
1/8-2X2 6063-T6
1/8-2X3 6063-T6
1/8-2X4 6063-T6
1/8-2X5 6063-T6
1/8-2X6 6063-T6



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PEDRO DE FIGUEIREDO
PE 52609

PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
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11600 ADELMO LANE
FORT MYERS, FL 33966
Tel.: (239) 278-3066 - Fax: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

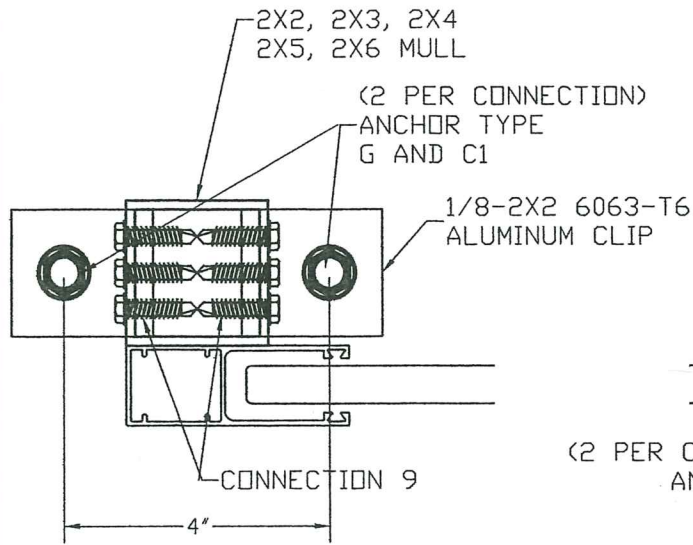
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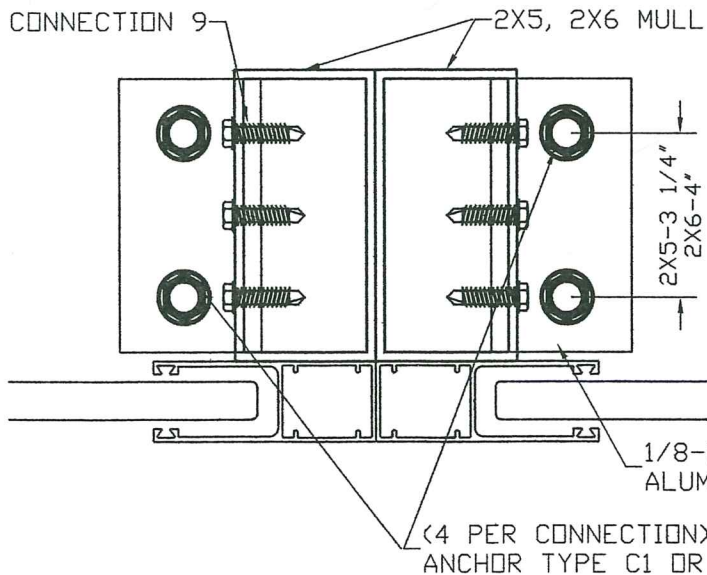
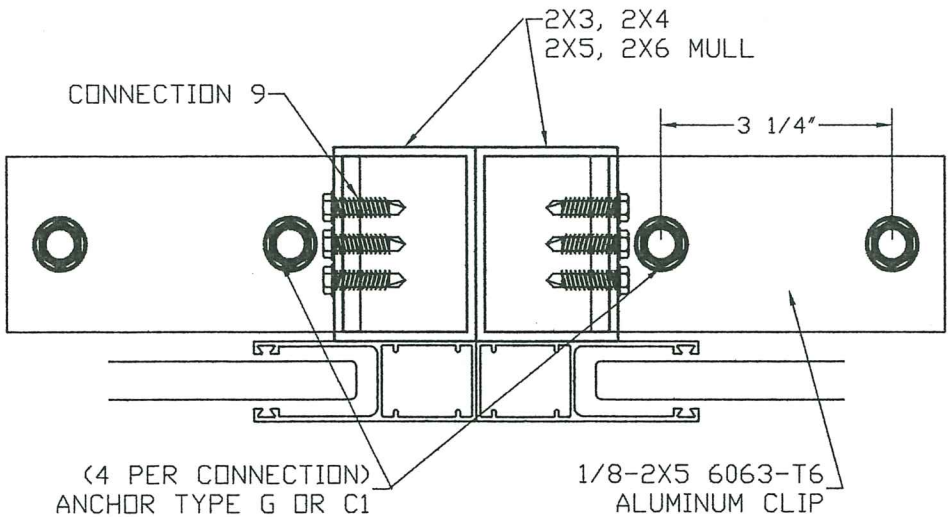
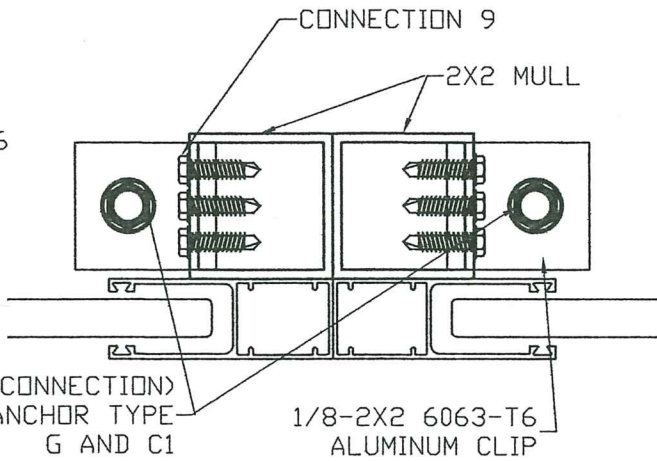
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MULLION BARS ANCHORAGE

CONNECTION 8
FLOOR/CEILING MOUNTED TO CONCRETE



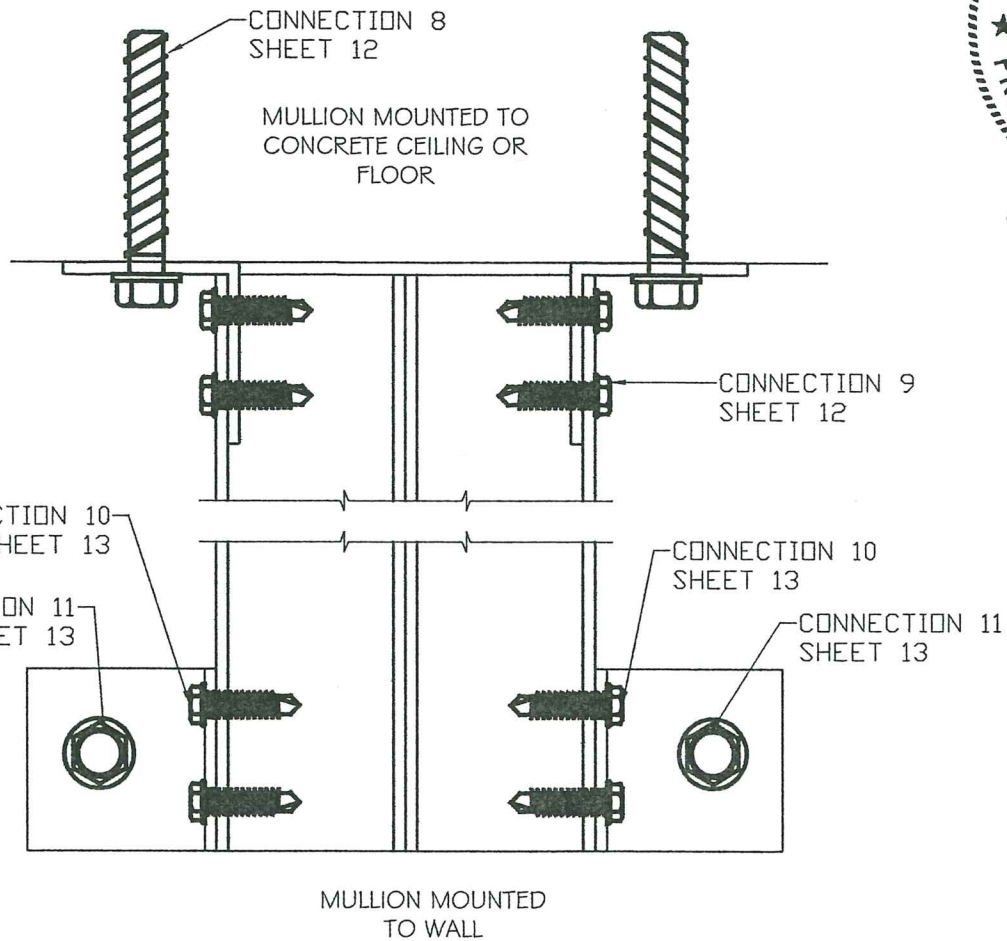
CONNECTION 8
APPLICABLE ONLY FOR
ANCHORS G AND C1



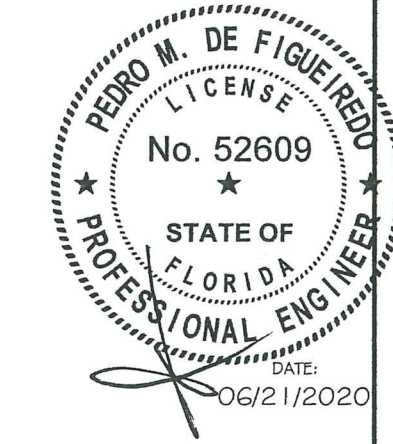
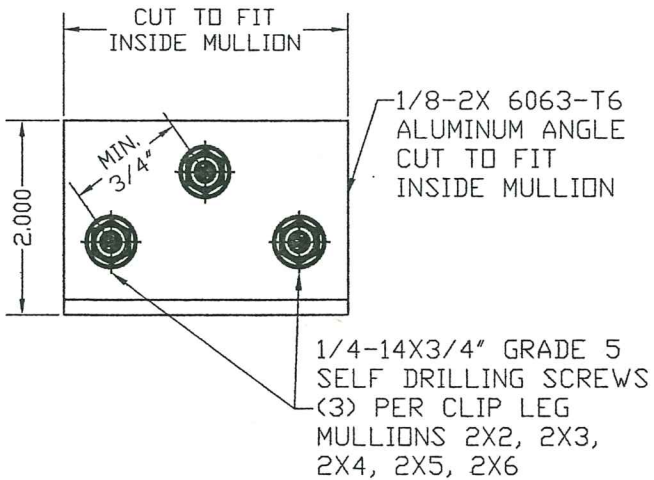
ANCHOR SCHEDULE

TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
C1	5/16" TAPCON BY ITW	2 1/4"	3"	3000 psi CONCRETE
G	3/8" WEDGE BOLTS BY POWERS	2 1/2"	4 1/2"	3000 psi CONCRETE

NOTES:
1- ED - EDGE DISTANCE
2- EMBEDMENT DIMENSION ARE TAKEN BEYOND THE WALL DRESSING



CONNECTION 9
CLIP ANCHORED TO MULLION BAR



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PRODUCT CATEGORY: SHUTTERS
PRODUCT MATERIAL: ALUMINUM
ES 55 SHUTTER
EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
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By *Healy A. Miller*
Miami Dade Product Control

MIAMI-DADE BCCO:

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CONNECTION 11

MULLION BARS ANCHORED TO WALL

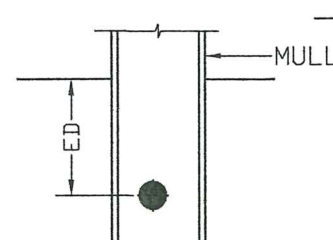
SINGLE MULLION CONNECTION 11 SELECTION

MULLION TYPE	TYPE OF ANCHORS			
	A	C2	C/C1/G	E
2x2	11.2	11.4	11.1	11.2
2x3	11.3	11.5	11.2	11.2
2x4	11.3	11.5	11.2	11.2
2x5	11.3	11.5	11.2	11.3
2x6	11.3	11.5	11.2	11.3

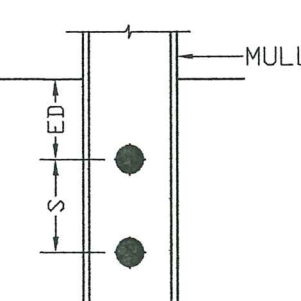
DUAL MULLIONS CONNECTION 11 SELECTION

MULLION TYPE	TYPE OF ANCHORS			
	A	C2	C/C1/G	E
2x2	11.7	11.9	11.6	11.7
2x3	11.8	11.10	11.7	11.7
2x4	11.8	11.10	11.7	11.7
2x5	11.8	11.10	11.7	11.8
2x6	11.8	11.10	11.7	11.8

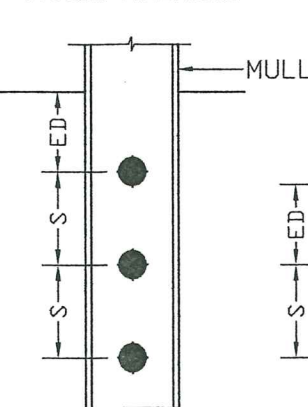
CONNECTION 11.1
SINGLE MULLION
ONE ANCHOR



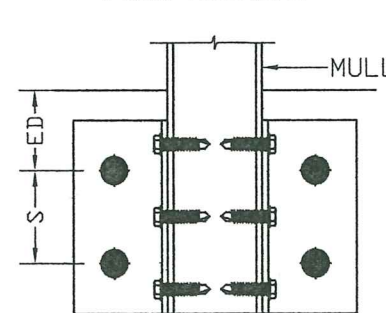
CONNECTION 11.2
SINGLE MULLION
TWO ANCHORS



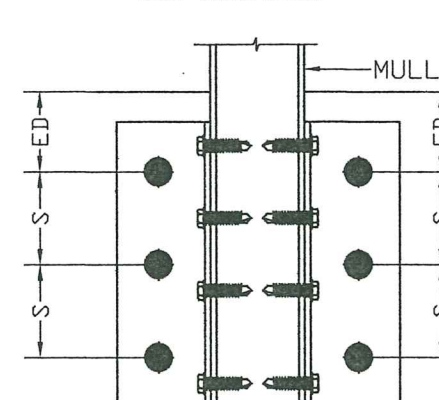
CONNECTION 11.3
SINGLE MULLION
THREE ANCHORS



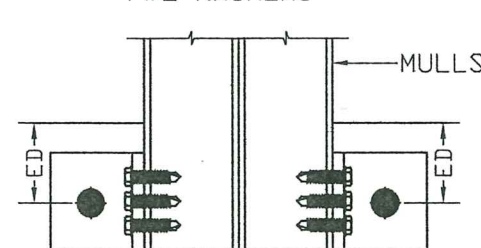
CONNECTION 11.4
SINGLE MULLION
FOUR ANCHORS



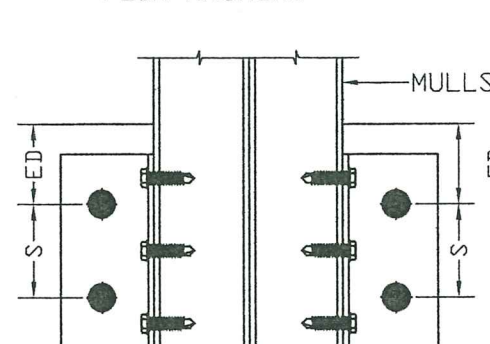
CONNECTION 11.5
SINGLE MULLION
SIX ANCHORS



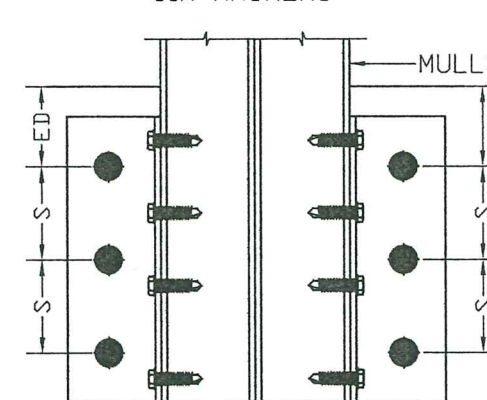
CONNECTION 11.6
DUAL MULLIONS
TWO ANCHORS



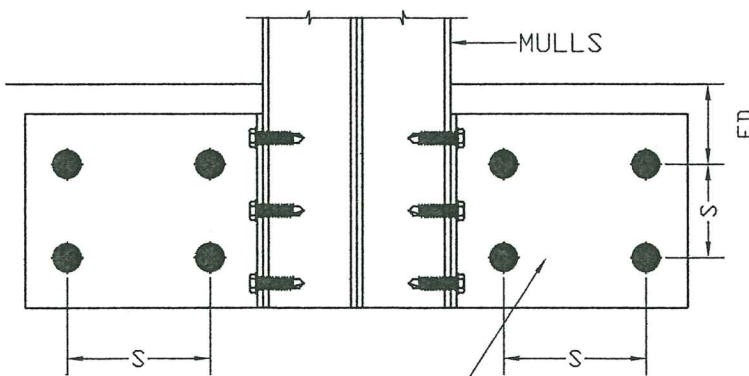
CONNECTION 11.7
DUAL MULLIONS
FOUR ANCHORS



CONNECTION 11.8
DUAL MULLIONS
SIX ANCHORS

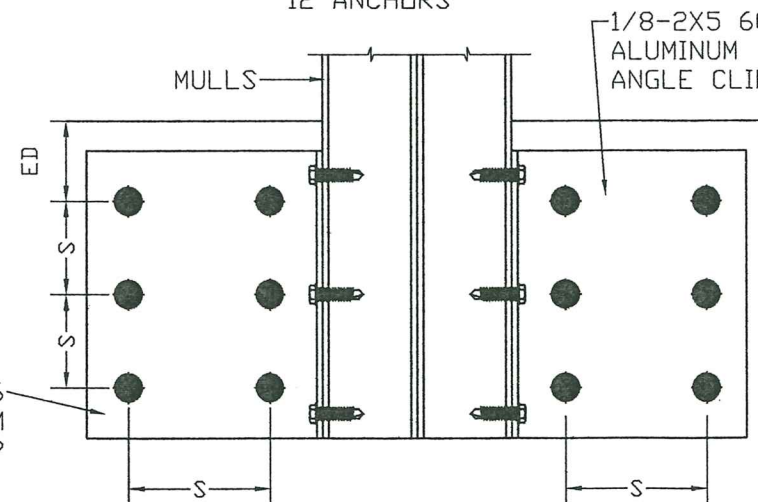


CONNECTION 11.9
DUAL MULLIONS
EIGHT ANCHORS



1/8-2X5 6063-T6
ALUMINUM
ANGLE CLIP

CONNECTION 11.10
DUAL MULLIONS
12 ANCHORS



1/8-2X5 6063-T6
ALUMINUM
ANGLE CLIP

ANCHOR SCHEDULE

TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
A	1/4" TAPCON BY ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
C2	5/16" TAPCON BY ITW	1 1/4"	4"	HOLLOW BLOCK
C	5/16" TAPCON BY ITW	2 1/4"	4"	GF BLOCK
C1	5/16" TAPCON BY ITW	2 1/4"	3"	3000 PSI CONCRETE
E	1/4" TAPCON/ LAGS BY ITW	1 3/4"	1"	WOOD (SG > .55)
G	3/8" WEDGE BOLTS BY POWERS	2 1/2"	3"	3000 psi CONCRETE

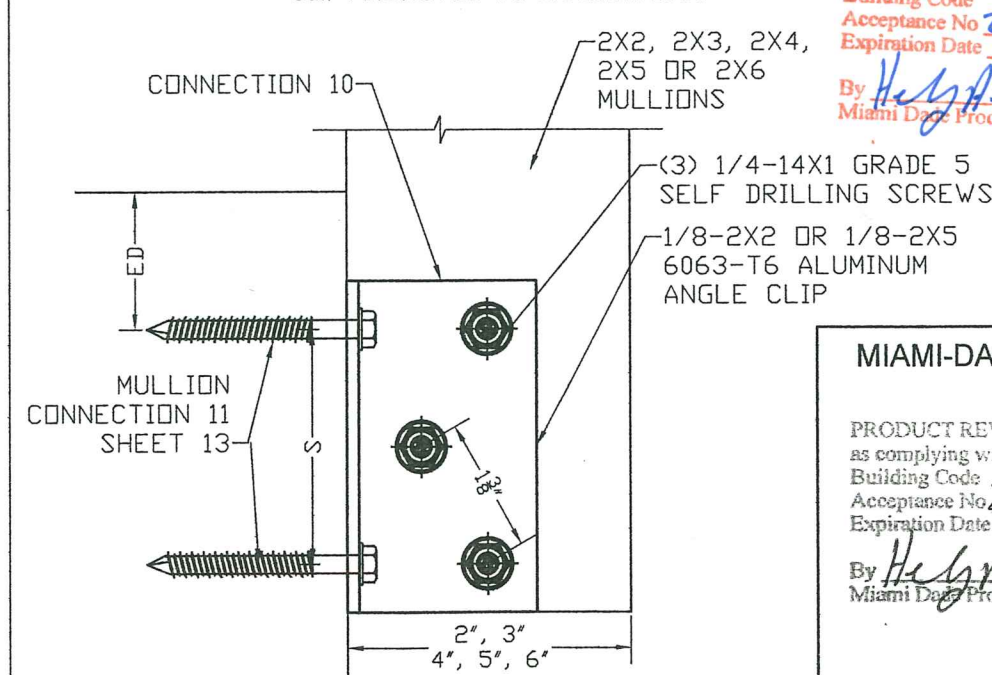
NOTE: 1- ED - EDGE DISTANCE
2- EMBEDMENT IS BEYOND THE WALL DRESSING

ANCHOR TYPE

	A	E	C,C1,C2	G
S	3.0	2.0	3.75	4.0

S - ANCHOR SPACING

CONNECTION 10
CLIP ANCHORED TO MULLION BAR



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Miami Dade Product Control

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By *Helga A. Miller*
Miami Dade Product Control



PRODUCT CATEGORY: SHUTTERS

PRODUCT MATERIAL: ALUMINUM

ES 55 SHUTTER

EXTRUDED ALUMINUM ROLL-UP SHUTTER

Manufacturer:
CROCI - North America

11600 ADELMO LANE
FORT MYERS, FL 33966
Tel.: (239) 278-3066 - Fax: (239) 278-5977
EMAIL: INFO@CROCIUSA.COM

SCALE:

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DATE:

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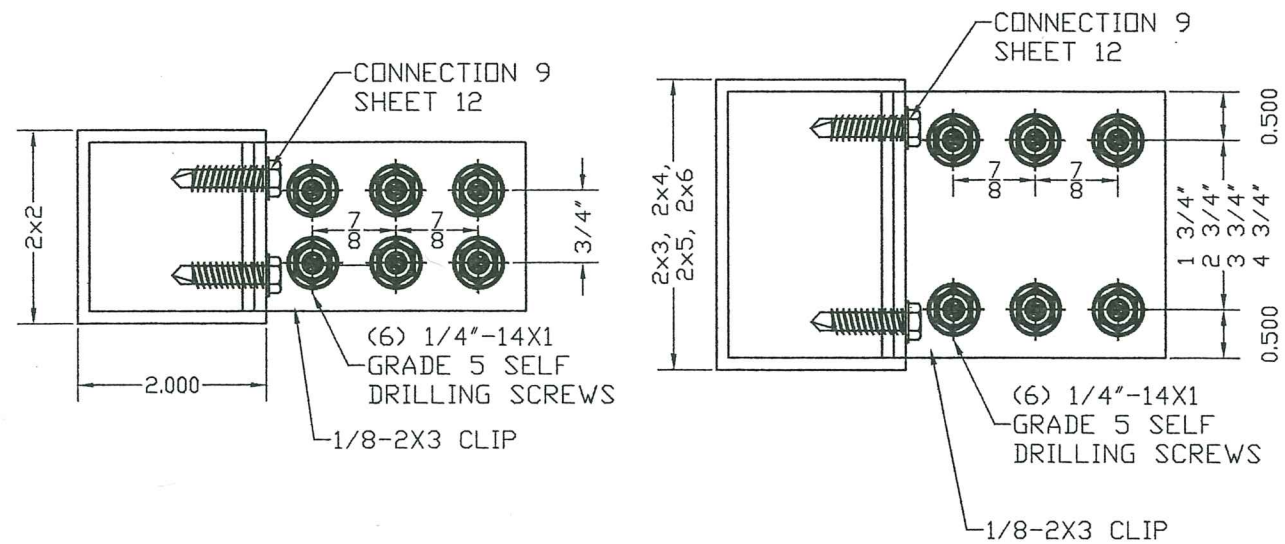
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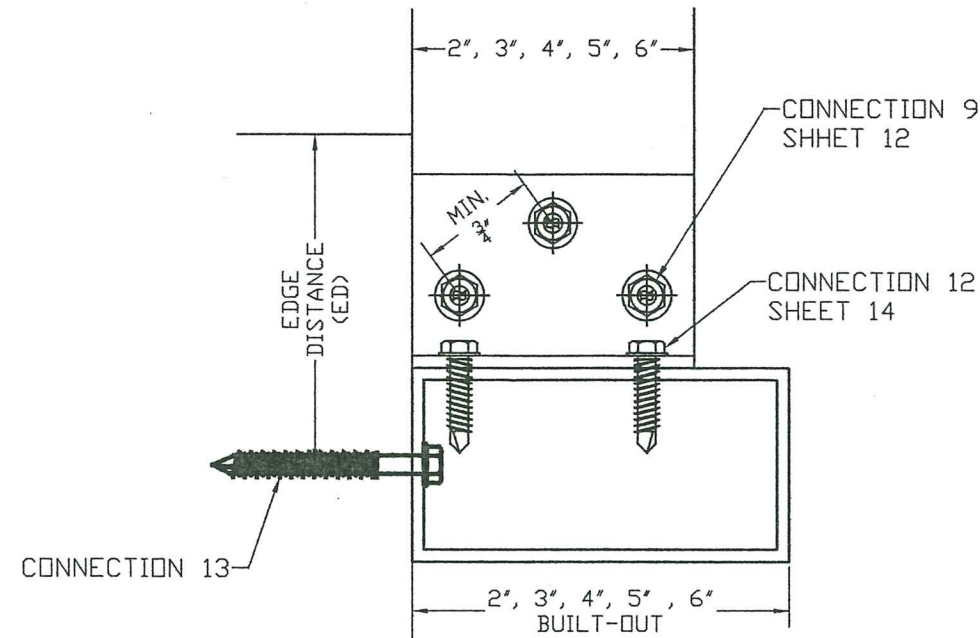


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CONNECTION 12 MULLION BARS ANCHORAGE FLOOR/CEILING TO BUILT-OUT TUBE



CONNECTION 13 BUILT-OUT WALL ANCHORAGE SUPPORTING MULLIONS



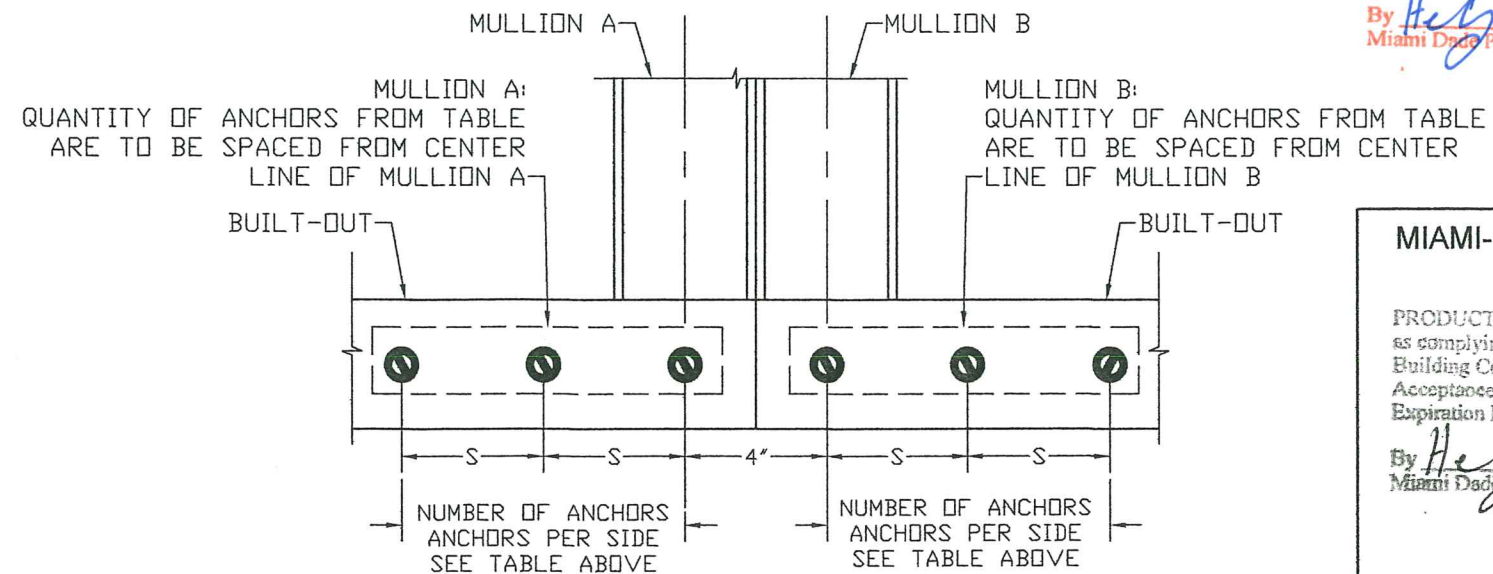
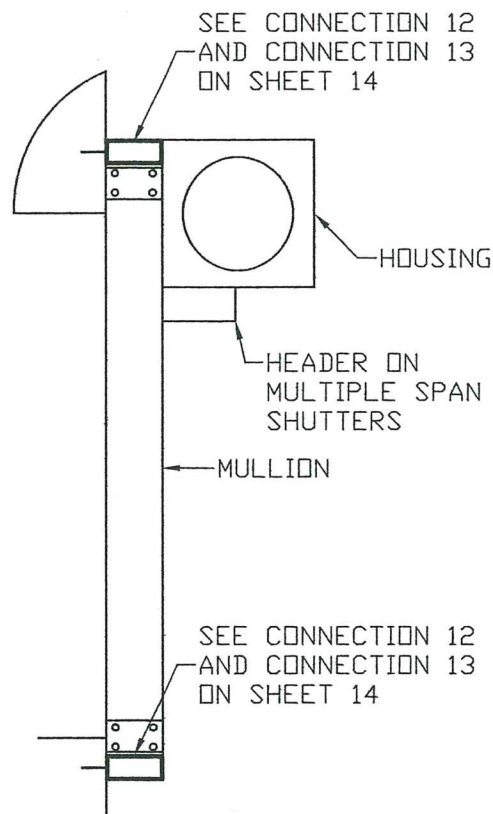
ANCHOR SCHEDULE

TYPE	DESCRIPTION	EMBEDMENT	ED	BASE MATERIAL
A	1/4\" TAPCON BY ITW	1 3/4"	2 1/2"	3000 psi CONCRETE
C2	5/16\" TAPCON BY ITW	1 1/4"	4"	HOLLOW BLOCK
C	5/16\" TAPCON BY ITW	2 1/4"	4"	GF BLOCK
C1	5/16\" TAPCON BY ITW	2 1/4"	3"	3000 PSI CONCRETE
E	1/4\" TAPCON/ LAGS BY ITW	1 3/4"	1"	WOOD (SG > .55)
G	3/8\" WEDGE BOLTS BY POWERS	2 1/2"	3"	3000 psi CONCRETE

NOTE: 1 - ED - EDGE DISTANCE
2- EMBEDMENT IS BEYOND THE WALL DRESSING

MINIMUM NUMBER OF ANCHORS PER BUILT-OUT

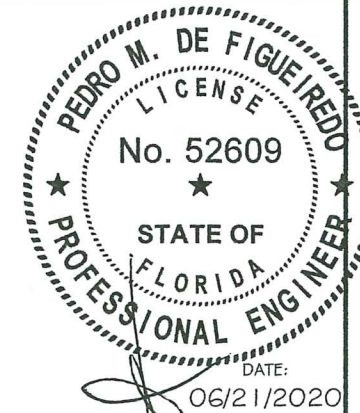
MULLION BARS	ANCHOR TYPE			
	A/E	C2	C1/C	G
2x2	3	7	2	2
2x3	5	11	3	2
2x4	5	12	4	3
2x5	5	12	4	3
2x6	5	12	4	3
S	3	3 3/4	3 3/4	4 1/2



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